Mathematical Architecture, Kinematic Calculations, and Interaction Mechanics of the Zeta Energy System

A Complete Walk-Through of the Hoard (2026i) Geometric Engine of Time, the Fifteen Lemmas, the $720 \to 72 \to 10$ Torsional Markov Pump, Monoid Collapse, Golden Strassen-Clifford Toroids, and Rational Trigonometric Replacements

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Abstract

This paper provides a complete, equation-by-equation walk-through of the calculations implemented in the upgraded `zeta_energy_system.html` simulation environment and integrates the complete deductive foundation of four companion monographs: *The Fifteen Lemmas of the Geometric Engine* (Hoard, October 16, 2026), *Monoid Collapse and Mass-Node Traversal in Relational Space*, *The Torsional Markov Memory-Offload Pump* (Hoard, June 21, 2026), and *Quaternionic Delta Encoding on Golden Strassen-Clifford Toroids*. Every computational subsystem is derived in detail: (1) the 30-wheel prime-weighted Dirichlet/Zeta signal proposer and its C_3 Fourier decomposition into an apex scalar invariant s and rotating plane z; (2) the projective Gaussian spinor clock $(1+i)^k$ in the binary octahedral group $2O$ and its square-root-free rational spread envelope $s_k = \sin^2(k\pi/4)$; (3) the joint 24-frame Chinese Remainder Theorem (CRT) schedule and its $720 \to 72 \to 10$ torsional gear nesting; (4) the embedding of the $\{4,3,5\}$ golden hyperbolic honeycomb cell, the corrected nine-phase tetrahedral wheel μ_9 in $\mathbb{Q}(2\cos 40^\circ)$, and the D_3-symmetric trefoil knot; (5) the seven distinct mathematical interaction mechanisms that govern trajectory collisions, inter-slice bridges, conformal null-cone inner products $\langle X_A, X_B \rangle$ in $\mathbb{R}^{4,1}$, and 64-bit Einstein Tile telemetry; and (6) the complete integration of all Fifteen Lemmas, Monoid Collapse, Golden Strassen-Clifford 16-MAC sub-cubic tensor execution, and exact algebraic replacements for $\sin, \cos, \tan, \sinh, \cosh, \tanh, \exp,$ and $\log$.

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1 System Architecture and Function-to-Theory Master Map

The Zeta Energy System models wave propagation, spinor phase synchronization, and hyperbolic compression across a $3 \times 3 \times 3$ cubic voxel lattice of spacing $C_{\text{space}} = 25.0$ units (core half-width $C_0 = 12.5$ units). Whereas the legacy 2025 version relied on heuristic floating-point trigonometric interpolation and treated frame 360 (Step 4) as a spatial singularity, the upgraded 2026i architecture grounds every calculation in the exact algebraic representatives of Hoard (2026i).

Table 1 maps every shader routine and JavaScript function in `zeta_energy_system.html` to the exact mathematical structure and theorem in the *Geometric Engine of Time* series that governs it.

Table 1: Master mapping between codebase functions, mathematical equations, and Geometric Engine of Time components.

Code Function / Shader	Mathematical Object	Role in Simulation	Engine Source
`getZetaPrimeWheel(t)` `getZeta(t)`	$Z(t) = \sum_{p \in P_{10}} p^{-1/2} \cos(t \ln p)$	Prime-weighted harmonic carrier wave across the 10 lowest wheel generators.	Hoard (2026e) Prop. 3.2; Prediction 2 (30-Wheel)
`vertexShaderTube`	$E_{\text{scalar}} = u_{\text{Scale}} G_{\text{env}}(s_k) \left(\frac{s}{3} + S(t)\right)$	Modulates tube radial expansion by the C_3 apex invariant $s/3$, soliton spikes, and spread s_k.	Prop. 2.3(3), Prop. 3.4, Prop. 8.2
`fragmentShaderTube` `fragmentShaderShell`	$l_j = \text{smoothstep}\left(\sin(\phi + \frac{2\pi j}{3}), \ldots\right)$	Renders the three 120° phase lobes and highlights $k \in \{2, 6\}$ handshake apices.	Prop. 2.1, Prop. 3.7, Prop. 7.1
`GAUSSIAN_CLOCK[k]`	$\epsilon_k = \frac{(1+i)^k}{2^{\lfloor k/2 \rfloor}} \in \{-1, 0, 1\}^2$	Exact 8-step projective spinor clock and rational spread $s_k \in \{0, \frac{1}{2}, 1\}$.	Prop. 2.2, Prop. 2.4 (Niven), Prop. 8.2
`applyWheelRotation40`	Circulant $R(\rho_0, \rho_1, \rho_2) \in \mathbb{Q}(2 \cos 40^\circ)$	Rotates the base cube triad Δ_0 by 40° and 80° about $(1, 1, 1)$ to form Δ_1, Δ_2.	Prop. 3.1 (Corrected Frame), Prop. 8.3
`getCoarseGearBlock`	$A_l = \{m : \lfloor 10m/72 \rfloor = l\}$	Maps fine frame $n \in [0, 720]$ to middle gear $m \in [0, 71]$ and coarse gear $l \in [0, 9]$.	Hoard (2026b); Prop. 3.9 (Pump)
`ExactTrefoilCurve`	$T(t) = (\sin t + 2 \sin 2t, \cos t - 2 \cos 2t, -\sin 3t)$	Embeds the D_3-symmetric knot inside the $\{4, 3, 5\}$ Klein cell ($r_{\text{max}} = 0.2520 < a$).	Prop. 3.11, Section 4.3

Code Function / Shader	Mathematical Object	Role in Simulation	Engine Source
`calculateNodePos`	$\mathbf{x}_i(n) \in \mathbb{R}^3$ for $i \in \{0..11\}$	Computes the 12-node trajectory via exact spinor-hyperbolic interpolation or legacy breath.	Prop. 2.3, Prop. 4.3, Prop. 4.4
`updateCompressionNodes`	$\mathbf{W}_{A,i}, \mathbf{W}_{B,i} \in \mathbb{R}^3$	Positions Slice A and Slice B in the $3 \times 3 \times 3$ grid and updates 3D inter-slice bridges.	Prop. 3.3, Section 4.3
`drawSpinorClockDial`	$(k_A, k_B) = (n \bmod 8, n_B \bmod 8)$	Interactive 8-node dial monitoring dual-slice spinor sheet lock and $\pm i$ handshakes.	Prop. 2.3(2)–(3), Prop. 3.7
`drawSlicer`	$d_i^2 = \|\mathbf{x}_{A,i} - \mathbf{x}_{B,i}\|^2$, $\langle X_A, X_B \rangle$	Detects 2D/3D trajectory hotspots, null-cone inner products, and 3×3 adjacent flux.	Prop. 3.3 (Axiom C2), Prop. 4.3
`updateCharts`	$s = a + b + c$, $\|z\|^2$, $E_{\text{bind}} = 2(ab+bc+ca)$	Real-time 12-channel spectrum of the C_3 Fourier split, quadrances, and cross-terms.	Prop. 3.4, Lemma 5 Part 6(a)
`updateQuaternionDisplay`	Form 3 64-bit Einstein Tile + Tracks A, B, C, R	Packs branch coordinates into 64 bits and verifies mod 881 residue fingerprints.	Def. 5.3, Prop. 8.7, Prop. 8.8, Def. 8.10

1.1 Integration of the Fifteen Lemmas and Four-Layer Deductive System into Section 1

By integrating *The Fifteen Lemmas of the Geometric Engine* (Hoard, October 16, 2026), the architectural table above is grounded in a strict **Four-Layer Deductive System plus Layer V (System Level)**:

1. **Origin-Centered Cubit Convention (Global Convention 1–5):** All spatial coordinates in `zeta_energy_system.html` are normalized to the standardized local acoustic cubit

$$C_0(f_{\text{cal}}; t, p, x_w, x_c) := \frac{\lambda}{16} = \frac{v_{\text{local}}(t, p, x_w, x_c)}{16 f_{\text{cal}}} = 53.748 \text{ cm} \pm 0.022 \text{ cm} \quad (\text{at } f_{\text{cal}} = 40 \text{ Hz}, 20^\circ\text{C}, 50\% \text{ RH}). \tag{1}$$

   In the simulation, `HALF_CUBE = 12.5` corresponds to $1\,C_0$, so the centered inner core is $K = [-C_0, C_0]^3$ (Cube 1, edge $2C_0 = \lambda/8$, containing the 8 core voxel centers) and the outer voxel cube is $V = [-2C_0, 2C_0]^3$ (Cube 2, edge $4C_0 = \lambda/4$, containing all $4 \times 4 \times 4 = 64$ voxel centers). As proven in Global Convention 4 and Lemma 5 Part 2, centering at the origin preserves the full 24-element octahedral rotation group $O \subset SO(3)$ and its 48-element spinor double cover $2O \subset S^3$ (which contains all 24 Hurwitz units $\mathcal{H}_1$, including

$h = \frac{1-i+j+k}{2}$), whereas a corner-anchored cube $[0, 4C_0]^3$ preserves only 3 rotations and breaks h.

2. **Mapping the 5 Toroidal Tiers and the $7.536\,C_0$ Separation Gap (Lemmas 4, 7, 10):** Beyond the Lower Manifold $V = [-2C_0, 2C_0]^3$ (which lies inside radius $\|x\| \le 2\sqrt{3}\,C_0 \approx 3.4641\,C_0$), the Upper Manifold $\mathcal{M}_{\text{upper}} = \bigcup_{n=0}^{4} T_n$ consists of five smoothly embedded tori with major radii $R_n = (5n^2 + 15n + 15)C_0 \in \{15, 35, 65, 105, 155\}C_0$, minor radii $r_n = (2n + 4)C_0 \in \{4, 6, 8, 10, 12\}C_0$, and integer windings $W_n = 16-2n \in \{16, 14, 12, 10, 8\}$, satisfying the exact invariants:

$$r_n+W_n = 20\,C_0 = \frac{5}{4}\lambda, \qquad \sum_{n=0}^{4} W_n = W_\Sigma = 60, \qquad \text{dist}(V, T_0) \ge (11-2\sqrt{3})C_0 \approx 7.5359\,C_0 \approx 4.050\ \text{m}. \tag{2}$$

By Lemma 7 Part 2, the central tier T_2 ($r_2 = 8, W_2 = 12$) is the unique arithmetic mean and median of both sequences.

3. **Synthetic Trigonometric Elimination at Layer I (Lemmas 1, 2, 3):** Before any shader or quaternion operation is called, Zhao's Three Lemmas eliminate inverse trigonometric calls ($\arccos, \arcsin, \text{atan2}$) from geometric setup:

   - *Lemma 1 (Isotomic Tangency Conservation):* Replaces angle-bisector tangent calculations on $\triangle ABC$ with linear semiperimeter differences $BD = CX = s - b$ and $BX = CD = s - c$, where $s = (a + b + c)/2$. This mirrors the mirror-symmetric tier radii r_n and r_{4-n} about $r_2 = 8C_0$.
   - *Lemma 2 (Spiral Similarity Center z_0):* Replaces polar angle/radius similarity solves between paired segments $AB \mapsto CD$ with a single complex rational division:

$$\alpha = \frac{c - d}{a - b}, \qquad z_0 = \frac{ad - bc}{a - b - c + d}. \tag{3}$$

   - *Lemma 3 (Symmedian Quadrance Ratio & Projective Harmonic Bundle):* Replaces the Law of Sines ($\sin\angle BAD/\sin\angle CAD = \sin C/\sin B$) with the exact ratio of adjacent squared lengths (quadrances $Q_3 = AB^2, Q_2 = AC^2$):

$$\frac{BF}{FC} = \frac{EB}{EC} = \frac{Q_3}{Q_2}, \qquad EA^2 = EB{\cdot}EC, \qquad (B, C; E, F) = -1, \qquad MB^2 = MC^2 = MF{\cdot}ME. \tag{4}$$

2 Signal Generation: The 30-Wheel Prime Proposer and C_3 Fourier Split

2.1 The Prime-Weighted Dirichlet/Zeta Wave Function

Each of the 9 longitudinal streams in the XY-grid (spanning $x, y \in \{-25, 0, +25\}$ and $z \in [-30, +30]$) is driven by a truncated Dirichlet-type harmonic series evaluated over the ten fundamental generator primes/units P_{10} associated with the 30-node primorial wheel (Hoard 2026e, Propositions 3.2 and 3.3; Hoard 2026i, Prediction 2):

$$P_{10} = \{1, 2, 3, 5, 7, 11, 13, 17, 19, 23\}. \tag{5}$$

In both the GLSL vertex shader (`getZetaPrimeWheel`) and the JavaScript spectrum engine (`getZeta`), the real carrier wave at phase coordinate t is computed as:

$$Z(t) = \sum_{p \in P_{10}} \frac{1}{\sqrt{p}} \cos(t \ln p). \tag{6}$$

For a point at normalized axial coordinate $u_x \in [0,1]$ along a tube of length $L_{\text{strip}} = 60$ at fine sequencer frame $n \in [0, 720]$, the local phase argument t is:

$$t(u_x, n) = 50u_x - 8\tau(n), \quad \text{where} \quad \tau(n) = \frac{n}{50}. \tag{7}$$

2.2 Threefold C_3 Branch Offsets and Soliton Spike Injection

By Proposition 2.1 (The Nine-Phase Factorization), the nine phases of μ_9 decompose into three $120°$ ($2\pi/3$) triads, which are cosets of $\mu_3 = \{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$. In the shader and spectrum analyzer, the three branches $j \in \{0, 1, 2\}$ of a stream are separated by exact $120°$ carrier shifts scaled by 10 rad^{-1}:

$$\Delta t_{120°} = 10 \cdot \frac{2\pi}{3} \approx 20.943951, \qquad \Delta t_{240°} = 10 \cdot \frac{4\pi}{3} \approx 41.887902. \tag{8}$$

Thus, at any axial slice t, the three real branch amplitudes (a, b, c) scaled by wave amplitude $A = u_{\text{Scale}}$ are:

$$a(t) = A \cdot Z(t), \qquad b(t) = A \cdot Z(t + 20.94395), \qquad c(t) = A \cdot Z(t + 41.88790). \tag{9}$$

When the user triggers `injectBtn` (*Inject Soliton Spike*), a Gaussian pulse timestamp $t_{\text{spike},m} = -\frac{8n}{50} - 5.0$ is pushed into the 10-slot uniform buffer `uSpikes[10]`, adding a localized soliton excitation:

$$S(t) = \sum_{m=0}^{9} 4 \exp\left(-0.2\,(t - t_{\text{spike},m})^2\right). \tag{10}$$

2.3 The C_3 Fourier Split and the 12 Spectrum Analysis Charts

Proposition 3.4 of Hoard (2026i) proves that the representation $\mathbb{R}^3$ of the cyclic group C_3 decomposes into a 1-dimensional fixed apex line and a 2-dimensional rotating plane ($\mathbb{R}^3 \cong \mathbb{R} \oplus \mathbb{R}^2$):

$$s = a + b + c \in \mathbb{R}, \qquad z = a + \omega b + \omega^2 c \in \mathbb{C}, \tag{11}$$

with exact inverse reconstruction:

$$a = \frac{1}{3}(s + z + \bar{z}), \qquad b = \frac{1}{3}(s + \omega^2 z + \omega \bar{z}), \qquad c = \frac{1}{3}(s + \omega z + \omega^2 \bar{z}). \tag{12}$$

Using the cyclotomic identity $1 + \omega + \omega^2 = 0$ (so $\omega + \omega^2 = -1$), the squared magnitude of the rotating plane vector z reduces purely to rational polynomial combinations of (a, b, c) without complex numbers or square roots:

$$\begin{aligned} |z|^2 &= (a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c) \\ &= a^2 + b^2 + c^2 + (\omega + \omega^2)(ab + bc + ca) \\ &= (a^2 + b^2 + c^2) - (ab + bc + ca). \end{aligned} \tag{13}$$

The function `updateCharts(time, amp)` evaluates 12 algebraic observables across 41 spatial samples $t \in [-8\tau, 50 - 8\tau]$ to populate Charts `c1`–`c12`:

1. **Charts `c1`, `c2`, `c3` (Branch Amplitudes):** $a(t)$, $b(t)$, and $c(t)$ from Eq. (9).
2. **Charts `c4`, `c5`, `c6` (Self-Quadrances):** $Q_0 = a^2$, $Q_1 = b^2$, and $Q_2 = c^2$.
3. **Charts `c7`, `c8`, `c9` (Pairwise Cross-Interaction Terms):**

$$I_{ab} = 2ab, \qquad I_{bc} = 2bc, \qquad I_{ca} = 2ca. \tag{14}$$

4. **Chart `c10` (C_3 Apex Scalar Invariant):** $s = a+b+c$ (fixed under cyclic branch permutation $(a,b,c) \mapsto (c,a,b)$).

5. **Chart `c11` (Rotating Plane Norm-Squared):** $|z|^2 = a^2+b^2+c^2-ab-bc-ca$ from Eq. (13).

6. **Chart `c12` (Inter-Lobe Binding Energy):** The interference excess between total squared apex amplitude $s^2 = (a+b+c)^2$ and the sum of individual branch quadrances:

$$E_{\text{bind}} = s^2 - (Q_0+Q_1+Q_2) = 2(ab+bc+ca) = \frac{2}{3}\left(s^2 - |z|^2\right). \tag{15}$$

2.4 Integration of Lemmas 4, 5, 11, and 14 & Trigonometric Replacements for Section 2

The signal-generation layer of Section 2 connects directly to four results in *The Fifteen Lemmas* and *Golden Strassen-Clifford Toroids*:

1. **Legendre Mod-5 and Garvan Mod-60 Partition of the 30-Node Wheel (Lemma 4 Part 5):** The $\phi(30) = 8$ prime-admissible unit residue classes $(\mathbb{Z}/30\mathbb{Z})^\times = \{1,7,11,13,17,19,23,29\}$ (encoded in Bits $[60:58]$ of the Einstein Tile) split into two 4-element cosets by the Legendre symbol $(b/5)$, which coincides with the level sets of χ_{60}^2 for Garvan's order-4 mock-theta Dirichlet character χ_{60}:

$$\mathcal{B}_+ = \ker\left(\frac{\cdot}{5}\right) = \{1,11,19,29\} \pmod{30}, \qquad \mathcal{B}_- = \{7,13,17,23\} \pmod{30}. \tag{16}$$

   Splitting the Zeta sum Eq. (6) into $Z_+(t)$ over $\mathcal{B}_+$ and $Z_-(t)$ over $\mathcal{B}_-$ provides a quadratic-residue parity channel before any primality test is run.

2. **Two-Product 128-Bit Lagrange Gram-TQF Replacement for Sine Area and Weitzenböck Ratio (Lemma 4 Part 2 & Lemma 5 Part 6a):** Given the three branch quadrances Q_1, Q_2, Q_3 (or 4D difference vectors $u, v \in \mathbb{Z}^4$ with $|u_k|, |v_k| \le 10^6 < 2^{20}$), standard geometry computes triangle area via $\frac{1}{2}\|u\|\|v\|\sin\theta$ or Heron's square-root formula. Lemma 5 Part 6(a) replaces both with Lagrange's overflow-safe 128-bit integer Gram identity:

$$A_R = 4\left(\|u\|^2\|v\|^2 - \langle u,v\rangle^2\right) = 4\left(Q_2Q_3 - \langle u,v\rangle^2\right) = (Q_1+Q_2+Q_3)^2 - 2\left(Q_1^2+Q_2^2+Q_3^2\right) < 2^{86}. \tag{17}$$

   By Lemma 4 Part 2, the algebraic identity

$$S^2 - 3A_R = 2\left[(Q_1-Q_2)^2 + (Q_2-Q_3)^2 + (Q_3-Q_1)^2\right] = 6\,|z|^2 \ge 0 \quad (S = Q_1+Q_2+Q_3) \tag{18}$$

   proves the quadrance form of Weitzenböck's inequality $\chi = \sqrt{A_R}/S \le 1/\sqrt{3}$, reaching its global maximum $\chi = 1/\sqrt{3}$ if and only if the three branch quadrances are balanced ($Q_1 = Q_2 = Q_3$).

3. **Golden Soliton Formation vs. the Fokker-Planck Circularity Boundary (Lemma 14 & Golden Strassen-Clifford Paper §4.1):** In *Quaternionic Delta Encoding on Golden Strassen-Clifford Toroids*, §4.1 models soliton formation by setting the diffusion tensor $D_2 \to 0$ when colliding streams land on a prime quadrance $Q = N(q) = p$. Lemma 14 Parts 2–4 formalize this mathematically: setting outgoing rates to zero on a target subset $\mathcal{P} \subset \Lambda$ causes probability mass to converge exponentially fast (at rate $\lambda_{\min} > 0$) to the **Harmonic Hitting Measure**:

$$H_{x_0}(p) = \left((-L_{TT})^{-1}L_{TP}\right)_{x_0,p}, \tag{19}$$

   and on the infinite 4D Hurwitz lattice $\mathcal{H}$, the Gibbs partition function $Z_{\mathcal{H}}(\beta) = 1 + \sum_{n=1}^{\infty} 24\sigma_{\text{odd}}(n)e^{-\beta V(n)}$ converges whenever $V(n) \ge \alpha \ln n$ with $\beta\alpha > 2$. By Lemma 14 Part 4 (the **Circularity Boundary**), because evaluating the trap indicator $\mathbf{1}_{\mathcal{P}}(Q)$ requires testing whether Q is prime, the Fokker-Planck wave operates in **Helix A** (candidate proposer/sampler) while **Helix B** certifies primality in $O(1)$ using the Form 1 bitmap or Miller–Rabin.

3 The Projective Spinor Clock, Spreads, and the 24-Frame Schedule

3.1 Why Projective Arithmetic is Mandatory: Niven's Theorem

A unit quaternion $q = \cos\alpha + u\sin\alpha$ rotates $\mathbb{R}^3$ via $x \mapsto qx\bar{q}$ by the spatial angle 2α about axis u, where α is the spinor angle (Proposition 2.3). To advance space by 90° per step requires a spinor angle of $\alpha = 45^\circ = \pi/4$, which has order 8.

By Proposition 2.4 (Niven's Bound), a rational unit quaternion of finite order can only have order $1, 2, 3, 4$, or 6, because $\cos(2\pi m/n) \in \mathbb{Q}$ occurs solely for $\cos\alpha \in \{0, \pm\frac{1}{2}, \pm 1\}$. Since $\cos 45^\circ = \sqrt{2}/2 \notin \mathbb{Q}$, no rational unit quaternion has order 8; the spinor step $(1+i)/\sqrt{2}$ lies in the binary octahedral group $2O$ (order 48) rather than the 24 Hurwitz units $2T$.

To eliminate $\sqrt{2}$ and floating-point drift, `zeta_energy_system.html` runs the clock projectively on the Gaussian integer $g = 1 + i$ (Proposition 2.2):

$$(1+i)^2 = 2i, \qquad (1+i)^4 = -4, \qquad (1+i)^8 = 16. \tag{20}$$

Dividing $(1+i)^k$ by the greatest common divisor of its real and imaginary parts, $\gcd = 2^{\lfloor k/2 \rfloor}$, yields the 8 reduced Gaussian states $\epsilon_k = a_k + ib_k \in \{-1, 0, 1\}^2$ stored in `GAUSSIAN_CLOCK` (Table 2).

Table 2: Exact projective Gaussian clock states, spatial rotations, and rational spreads implemented in `GAUSSIAN_CLOCK`.

Step k	**Stage Name**	**Unreduced** $(1+i)^k$	$\gcd = 2^{\lfloor k/2 \rfloor}$	**Reduced** ϵ_k	**Spinor** q_k	**Spread** s_k
0	Zero Point	1	1	$1 + 0i$	$+1$	0 (Node)
1	Seed	$1 + i$	1	$1 + i$	$(1+i)/\sqrt{2}$	$1/2$
2	Onload	$2i$	2	$0 + i$	$+i$ (Apex)	1 (Max)
3	Project	$-2 + 2i$	2	$-1 + i$	$(-1+i)/\sqrt{2}$	$1/2$
4	Invert	-4	4	$-1 + 0i$	-1 (Sheet)	0 (Node)
5	Fold	$-4 - 4i$	4	$-1 - i$	$(-1-i)/\sqrt{2}$	$1/2$
6	Offload	$-8i$	8	$0 - i$	$-i$ (Apex)	1 (Max)
7	Index	$8 - 8i$	8	$1 - i$	$(1-i)/\sqrt{2}$	$1/2$

3.2 Rational Spreads Without Square Roots

By Proposition 8.2, the transverse envelope $A_k = |\sin(k\pi/4)|$ is represented exactly by its rational spread:

$$s_k = \frac{b_k^2}{a_k^2 + b_k^2} = \sin^2\left(\frac{k\pi}{4}\right) \in \left\{0, \frac{1}{2}, 1\right\}, \qquad c_k = \frac{a_k^2}{a_k^2 + b_k^2} = 1 - s_k. \tag{21}$$

In `vertexShaderTube`, the uniform `uSpread` passes s_k directly to the GPU to modulate the scalar energy envelope and radial normal displacement:

$$G_{\text{env}}(s_k) = 0.65 + 0.35 s_k, \qquad E_{\text{scalar}} = u_{\text{Scale}} \cdot G_{\text{env}}(s_k) \left(\frac{p_1 + p_2 + p_3}{3} + S(t)\right), \tag{22}$$

$$\mathbf{p}_{\text{displaced}} = \mathbf{p} + 0.11\,\mathbf{n}\,E_{\text{scalar}}\,(1 + 0.4\,u_{\text{GoldenBoost}})\,. \tag{23}$$

Notice how this corrects the legacy 2025 error (Hoard 2026i, Section 10.2 Errata #3): Step $k = 4$ has spread $s_4 = 0$ (an envelope node on the negative spinor sheet $q_4 = -1$), whereas Steps $k = 2$ ($+i$) and $k = 6$ ($-i$) are the true handshake apices where $s_2 = s_6 = 1$.

3.3 The Joint 24-Frame CRT Schedule and $C_3 \times C_2$ Handshakes

In `updateTelemetryBar(n)`, the sequencer advances the spinor clock and the branch index simultaneously once per frame (Definition 3.6 and Proposition 3.7):

$$k(n) = n \bmod 8, \qquad j(n) = n \bmod 3. \tag{24}$$

Because $\gcd(8, 3) = 1$, the Chinese Remainder Theorem yields the group isomorphism:

$$\mathbb{Z}/24\mathbb{Z} \xrightarrow{\sim} \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}, \qquad n \bmod 24 \mapsto (n \bmod 8,\ n \bmod 3). \tag{25}$$

Within each 24-frame cycle, the six handshake events ($k \in \{2, 6\}$, where $s_k = 1$) occur at frames $n = 4m + 2$ for $m \in \{0, 1, 2, 3, 4, 5\}$ (Proposition 7.1):

- **Onload Windows ($k = 2$, Spinor $+i$):** Frame $n \equiv 2$ (Branch $j = 2$), Frame $n \equiv 10$ (Branch $j = 1$), and Frame $n \equiv 18$ (Branch $j = 0$).
- **Offload Windows ($k = 6$, Spinor $-i$):** Frame $n \equiv 6$ (Branch $j = 0$), Frame $n \equiv 14$ (Branch $j = 2$), and Frame $n \equiv 22$ (Branch $j = 1$).

Every branch j reaches its offload window exactly $\Delta = 12$ frames after its onload window, as $\Delta = 12$ is the unique solution modulo 24 to $\Delta \equiv 4 \pmod 8$ and $\Delta \equiv 0 \pmod 3$.

3.4 Integration of Lemmas 4, 5, 11, and 13 & Trigonometric Replacements for Section 3

Integrating *The Fifteen Lemmas* enriches the spinor clock and scheduling layer with four exact algebraic mechanisms:

1. **Impossibility of Projective $45°$ Spatial Rotations vs. Exact $90°$ Steps (Lemma 13 Part 4):** Why does the Engine step space by $90°$ ($\alpha = 45°$ spinor angle) rather than $45°$ ($\alpha = 22.5°$ spinor angle)? Lemma 13 Part 4 proves that for any projective integer quaternion $Q = w + \mathbf{v} \in \mathbb{Z}^4$ rotating $\mathbb{R}^3$ by spatial angle 2θ, the ratio $\|\mathbf{v}\|^2/w^2 = \tan^2\theta$ must be rational. For a $45°$ spatial rotation ($\theta = 22.5°$),

   $$\tan^2(22.5°) = (\sqrt{2} - 1)^2 = 3 - 2\sqrt{2} \notin \mathbb{Q}, \tag{26}$$

   proving that a $45°$ spatial step is impossible even projectively over $\mathbb{Z}^4$, whereas $\theta = 45°$ ($90°$ spatial step) has $\tan^2(45°) = 1 \in \mathbb{Q}$, realized by $Q_{\text{step}} = 1 + i$.

2. **Exact Inter-Tier Signed-Permutation Matrices $U_{\text{even}}, U_{\text{odd}}$ and Hurwitz Spinor h (Lemma 4 Parts 2 & 4, Prop. 7.12):** Across the 5 toroidal tiers $T_0, \ldots, T_4$, alternating orientations $\text{Rot}_x(\pi/2)$ and $\text{Rot}_y(\pi/2)$ compose to the exact integer signed-permutation matrices:

   $$U_{\text{even}} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{pmatrix}, \qquad U_{\text{odd}} = U_{\text{even}}^{-1} = \begin{pmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \tag{27}$$

   lifted to S^3 by the order-6 Hurwitz unit $h = \frac{1-i+j+k}{2}$ ($h^2 = -\bar{h}, h^3 = -1, h^4 = -h, h^5 = \bar{h}, h^6 = +1$). By Proposition 7.12, carrying data out from the core through $T_0 \to T_4$ via $W = U_3U_2U_1U_0$ and offloading back from $T_4 \to T_0$ satisfies $W = I$ and $h\bar{h}h\bar{h} = 1$ with rational radial scale $R_4/R_0 = 155/15 = 31/3$ out and $3/31$ back, and $N(hq) = N(q)$ (zero bit growth).

3. **Harmonic Bundle Gate Lengths** $(0, 2C_0; 4C_0, \frac{4}{3}C_0) = -1$ **(Lemma 5 Part 3):** Along the core-to-cube baseline $B = 0, C = 2C_0, E = 4C_0$, the symmedian foot of the Apollonius circle of radius $R_A = 2\sqrt{2}\,C_0$ defines the **Klein-Wave Merge Boundary** $x_F = \frac{4}{3}C_0 = \frac{\lambda}{12}$ (acoustic phase $\pi/6$, quadrance ratio $Q_3/Q_2 = 2$) and radial half-edge conjugate $F_{\text{rad}} = \frac{2}{3}C_0 = \frac{\lambda}{24}$ (phase $\pi/12$), encoded directly in Bits $[49 : 48]$ of the Einstein Tile Gate Byte.

4. **Steinhaus–Sós Three-Distance Theorem & Fibonacci Two-Gap Perfection (Lemma 11 Part 3):** When the Scale Register (Bits $[35 : 32]$) selects Fibonacci index m ($N = F_m \in \{5, 8, 13, 21, 34, 55, 89\}$), golden circle subdivision $\{k/\varphi \bmod 1 : 0 \le k < F_m\}$ partitions the circle into F_{m-1} gaps of length $\varphi^{-(m-2)}$ and F_{m-2} gaps of length $\varphi^{-(m-1)}$ (in exact ratio $\varphi : 1$) with zero collisions.

4 The $720 \to 72 \to 10$ Torsional Markov Pump and Multi-Clock Closures

4.1 Three-Gear Block Telescoping and Open Holonomy

The 720-frame master cycle nests 90 spinor cycles ($720 = 90 \times 8$) and 30 joint CRT cycles ($720 = 30 \times 24$). In `updateTelemetryBar(n)` and `getCoarseGearBlock(m)`, the frame index $n \in [0, 719]$ drives the three-scale torsional pump of Hoard (2026b) and Proposition 3.9:

1. **Outer 12-Point Fine Gear (**720 **frames):** Tracks fine state increments $\Delta q_n = q_n q_{n-1}^{-1}$ with open boundary convention $q_{-1} = q_0$.

2. **Middle 10-Point Aggregate Gear (**72 **blocks):** Indexed by $m = \lfloor n/10 \rfloor \in \{0, \ldots, 71\}$. Each 10-frame block telescopes to its boundary ratio:

$$\Delta Q_m = \Delta q_{10m+9} \Delta q_{10m+8} \cdots \Delta q_{10m} = q_{10m+9} q_{10m-1}^{-1}, \qquad \delta_m = \Delta Q_m^{-1}. \tag{28}$$

3. **Inner 8-Point Coarse Gear (**10 **blocks):** Indexed by

$$l(m) = \left\lfloor \frac{10m}{72} \right\rfloor \in \{0, \ldots, 9\}, \tag{29}$$

   which partitions the 72 middle blocks into ten contiguous runs $A_l = \{m : \lfloor 10m/72 \rfloor = l\}$ of exact sizes:

$$(|A_0|, |A_1|, \ldots, |A_9|) = (8, 7, 7, 7, 7, 8, 7, 7, 7, 7), \qquad \sum_{l=0}^{9} |A_l| = 72. \tag{30}$$

By Proposition 3.9(2), the right-acting middle product $Q^{(10)} = \delta_0 \delta_1 \cdots \delta_{71} = P^{-1}$ and coarse product $Q^{(8)} = \prod_{l=0}^{9} \left(\prod_{m \in A_l} \delta_m \right) = P^{-1}$ combine with the open product $P = q_{719} q_0^{-1}$ to yield the pump holonomy:

$$H = Q^{(8)} Q^{(10)} P = P^{-1} P^{-1} P = P^{-1}. \tag{31}$$

As proven in Proposition 3.13, if the first increment were defined cyclically ($\Delta q_0 = q_0 q_{719}^{-1}$), the product would telescope to 1 identically for every sequence of states regardless of numerical drift. The telemetry bar therefore tracks the **open holonomy** $P = q_n q_0^{-1}$ (`#tele-holonomy`), confirming closure ($P = +1$) only when $k = n \bmod 8 = 0$.

4.2 Physical Frequency Closure Windows

Under Definition 3.8 (1 frame $= \frac{1}{60}$ s), the telemetry pill `#tele-closures` evaluates five simultaneous modular divisibility conditions at every frame n (Proposition 3.8 and Table 11):

$$\text{Caesium-133 } (9,192,631,770 \text{ Hz}): \quad \frac{\nu_{\mathsf{Cs}}}{60} n = 153,210,529.5\, n \in \mathbb{Z} \iff 2 \mid n, \tag{32}$$

$$\text{40 Hz Carrier}: \quad \frac{40}{60} n = \frac{2n}{3} \in \mathbb{Z} \iff 3 \mid n, \tag{33}$$

$$\text{432 Hz Carrier}: \quad \frac{432}{60} n = \frac{36n}{5} \in \mathbb{Z} \iff 5 \mid n, \tag{34}$$

$$\text{Joint 24-Frame Register}: \quad 24 \mid n, \tag{35}$$

$$\text{Master Synchronized Closure}: \quad \mathrm{lcm}(24,2,3,5) = 120 \mid n \quad (6 \text{ closures per } 720 \text{ frames}). \tag{36}$$

4.3 Integration of the Torsional Markov Pump Paper, Lemma 7, and Lemma 12 into Section 4

Combining *The Torsional Markov Memory-Offload Pump* (Hoard, June 21, 2026) with Lemma 7 and Lemma 12 of *The Fifteen Lemmas* yields four critical mathematical upgrades and transcendental replacements:

1. **Why Right-Acting Middle Multiplication is Load-Bearing (Lemma 12 Part 2):** In the June 2026 Pump paper (§11), the middle net rotation was written schematically as $Q^{(10)} = \prod_{m=0}^{71} \Delta q_m^{(10)}$. Because quaternion multiplication is non-commutative and inversion reverses product order ($(AB)^{-1} = B^{-1}A^{-1}$), Lemma 12 Part 2 proves that while the outer gear multiplies left-to-right in reverse chronological order $Q^{(12)} = \Delta Q_{71}^{(12\to10)} \cdots \Delta Q_0^{(12\to10)}$, the middle and inner gears **must multiply in right-acting chronological order**:

$$Q^{(10)} = \delta_0\delta_1\cdots\delta_{71} = \left(\Delta Q_{71}^{(12\to10)} \cdots \Delta Q_0^{(12\to10)}\right)^{-1} = \left(Q^{(12)}\right)^{-1}, \qquad Q^{(8)} = \Delta q_0^{(8)} \cdots \Delta q_9^{(8)} = Q^{(10)}. \tag{37}$$

   If $Q^{(10)}$ were multiplied in left-acting order $\delta_{71}\cdots\delta_0$, $Q^{(10)}Q^{(12)} \neq 1$ for generic non-commuting quaternions.

2. **Replacing Transcendental SLERP $q^w = \exp(w \log q)$ with Rational Cayley Tangential Lifts (Pump Paper §9 & §14 vs. Lemma 9 Part 3):** In §9 of the Pump paper, upward reconstruction from inner to middle ($\Delta_c q_m^{(10)} = (\Delta q_{l(m)}^{(8)})^{w_m}$) and middle to outer ($\Delta_c q_n^{(12)} = (\Delta q_{m(n)}^{(10)})^{1/10}$) uses the transcendental SLERP formula $q^w = \exp(w \log q)$, which §14 quarantines from the executable forward path. To bring upward reconstruction inside the exact rational kernel without $\exp, \log, \arccos$, or $\sin$:

   - Represent any principal-branch unit quaternion $q = (w_0, \mathbf{m}) \in S^3 \cap \mathbb{Q}^4$ ($w_0 > 0$) by its rational Cayley vector $\mathbf{u} = \frac{\mathbf{m}}{1+w_0} = \tan\left(\frac{\theta}{2}\right)\hat{\mathbf{u}} \in \mathbb{Q}^3$ (inverse of $\mathcal{C}_{\mathsf{rat}}(\mathbf{u}) = \frac{(1-\|\mathbf{u}\|^2, 2\mathbf{u})}{1+\|\mathbf{u}\|^2}$, Lemma 9 Part 3).
   - For small or rational-approximated fractional steps $w \in \mathbb{Q}$ (such as $w = 1/10$), apply the rational tangent-addition / Padé approximant $\mathbf{u}_w = w\,\mathbf{u}\left(1 + \frac{1-w^2}{3}\|\mathbf{u}\|^2\right)^{-1} \in \mathbb{Q}^3$ and map back via $\mathcal{C}_{\mathsf{rat}}(\mathbf{u}_w) \in S^3 \cap \mathbb{Q}^4$. Every lifted increment $\hat{q}_n^{(12)}$ then has exact unit norm $N(\mathcal{C}_{\mathsf{rat}}(\mathbf{u}_w)) = 1$ over $\mathbb{Q}$ with zero floating-point renormalization!

3. **Replacing $\epsilon = \log H$ Drift Redistribution with BCH Cross-Products (Pump Paper §11 vs. Lemma 7 Part 3b):** In §11 of the Pump paper, holonomy drift $\epsilon = \log H$ is redistributed across the three gears via $\Delta q_k^{(r)} \leftarrow \exp(-\lambda_r \epsilon)\Delta q_k^{(r)}$ ($\lambda_{12} + \lambda_{10} + \lambda_8 = 1$). By Lemma 7 Part

3(b), for three drift quaternions $H_a = \exp(\epsilon_a)$ ($\epsilon_a \in \mathfrak{su}(2) \cong \mathbb{R}^3$), the compensating inverse $H_4 = (H_1H_2H_3)^{-1} = \bar{H}_3\bar{H}_2\bar{H}_1$ has the Baker–Campbell–Hausdorff expansion:

$$-\log H_4 = \epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_1 \times \epsilon_2 + \epsilon_1 \times \epsilon_3 + \epsilon_2 \times \epsilon_3 + \mathcal{O}(\|\epsilon\|^3), \tag{38}$$

allowing non-abelian commutator drift across the three gears to be computed via pure 3D vector cross products and mapped back to $S^3 \cap \mathbb{Q}^4$ via $\mathcal{C}_{\mathsf{rat}}(-\frac{1}{2}\lambda_r\epsilon)$.

4. **Exact Rational Valve Kernel for Positive Markov Circulation $J > 0$ (Pump Paper §10 vs. Lemma 7 Part 3c & 3e):** Instead of using transcendental exponential rates $e^{-\beta\Delta V}$ to break detailed balance on the gear states $\{12, 10, 8\}$, Lemma 7 Part 3(c) takes a symmetric base walk $P^0_{ij} = W_{ij}/d_i$ and multiplies each reverse edge $e \in \{(10, 12), (8, 10), (12, 8)\}$ by the exact rational valve factor:

$$\rho_e = \frac{1}{1 + \lambda A_e} \in \mathbb{Q} \cap (0, 1], \qquad J = \frac{(1 - \rho_{10,12}\rho_{8,10}\rho_{12,8})\Pi_+}{Z} > 0. \tag{39}$$

On the 64-voxel torus $\Lambda = (\mathbb{Z}/4\mathbb{Z})^3$ (Lemma 7 Part 3e), translation-invariant rational valves $\rho_k \in (0, 1)$ yield the exact uniform stationary distribution $\pi = 1/64$ and positive forward edge current $J_{x\to x+e_k} = \frac{(1-\rho_k)w_k}{64W} > 0$.

5 3D Geometric Substructures and Node Kinematics

5.1 The $\{4, 3, 5\}$ Golden Hyperbolic Honeycomb Cell

Inside the central core cube $K = [-C_0, C_0]^3$ ($C_0 = 12.5$), `goldenCellGroup` renders the central cell $[-aC_0, aC_0]^3$ of the compact hyperbolic $\{4, 3, 5\}$ honeycomb in the Klein projective model (Proposition 4.3 and Proposition 8.4):

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.61803399, \qquad a = \varphi^{-3/2} = \sqrt{\sqrt{5} - 2} \approx 0.48586827. \tag{40}$$

Its hyperbolic inradius r_{in} and face-neighbour Lorentz boost rapidity $\vartheta = 2r_{\mathsf{in}}$ satisfy:

$$r_{\mathsf{in}} = \mathsf{artanh}(a) \approx 0.53063753, \qquad \cosh\vartheta = \varphi, \qquad \sinh\vartheta = \sqrt{\varphi}, \qquad \sinh^2(r_{\mathsf{in}}) = \frac{1}{2\varphi} = \cos 72^\circ. \tag{41}$$

5.2 The Corrected Nine-Phase Tetrahedral Wheel μ_9

In `tetraWheelGroup`, three regular tetrahedra $\Delta_0, \Delta_1, \Delta_2$ of side $2\sqrt{2}\,C_0$ share the common apex $\mathbf{x}_* = C_0(1, 1, 1)$. Following the frame correction of Proposition 3.1(4), the $a = 0$ triad lies on the three core cube vertices whose coordinates sum to $-C_0$:

$$\mathbf{v}_{0,0} = C_0(1, -1, -1), \qquad \mathbf{v}_{0,1} = C_0(-1, 1, -1), \qquad \mathbf{v}_{0,2} = C_0(-1, -1, 1). \tag{42}$$

To generate the $a = 1$ ($+40^\circ$) and $a = 2$ ($+80^\circ$) tetrahedra, the function `applyWheelRotation40(vec, a)` applies the exact circulant rotation matrix $R \in \mathbb{Q}(c)$ where $c = 2\cos 40^\circ$ is a root of $c^3 - 3c + 1 = 0$ (Proposition 8.3):

$$R = \begin{pmatrix} \rho_0 & \rho_1 & \rho_2 \\ \rho_2 & \rho_0 & \rho_1 \\ \rho_1 & \rho_2 & \rho_0 \end{pmatrix}, \qquad \rho_0 = \frac{1 + c}{3}, \quad \rho_1 = \frac{3 - c - c^2}{3}, \quad \rho_2 = \frac{c^2 - 1}{3}, \tag{43}$$

where ρ_0, ρ_1, ρ_2 are the three roots of $9y^3 - 9y^2 + 1 = 0$, satisfying $R^TR = I$, $R^3 = g_3$ (the 120° coordinate permutation), and $R^9 = I$ exactly.

5.3 Node Kinematics in `calculateNodePos(nodeIndex, step)`

Each slice tracks 12 nodes $i \in \{0, \ldots, 11\}$ partitioned into three 4-node branch groups ($i \in \{0..3\}$ Red, $i \in \{4..7\}$ Green, $i \in \{8..11\}$ Blue). Let $\mathbf{e}_i$ denote the 12 midpoints of the cube edges of half-width $C_0 = 12.5$, and let $\mathbf{f}_{f(i)}$ denote the 6 cube face centers where $f(i) = 2\lfloor i/4 \rfloor + (i \bmod 2) \in \{0, \ldots, 5\}$. The golden Klein face centers are $\mathbf{g}_{f(i)} = a\,\mathbf{f}_{f(i)} = \varphi^{-3/2}\mathbf{f}_{f(i)}$.

In the default **2026i Exact Spinor-Hyperbolic Mode** (`exact2026`), the position $\mathbf{x}_i(n)$ at frame $n \in [0, 720]$ is governed by three coupled equations:

1. **Macro-Gear Angle and Transverse Spread Envelope:**

$$\Phi(n) = \frac{2\pi n}{720}, \qquad \kappa(n) = \frac{n}{90} \in [0, 8], \qquad s_{\text{env}}(n) = \sin^2\!\left(\frac{\kappa(n)\pi}{4}\right). \tag{44}$$

   Notice that $s_{\text{env}}(n)$ equals 1 at the macro-handshake frames $n = 180$ ($\kappa = 2$) and $n = 540$ ($\kappa = 6$), and vanishes (0) at $n = 0, 360, 720$ ($\kappa = 0, 4, 8$).

2. **Hyperbolic Klein Radius Modulation:** The radial scale interpolates smoothly between the scaled trefoil knot's maximum Klein radius $r_{\text{trefoil}} = 0.2520$ (Hoard 2026i, Section 4.3), the golden cell boundary $a = 0.4859$, and the unit core cube 1.0:

$$r_{\text{Klein}}(n) = 0.2520 + (1 - 0.2520)\,(0.35 + 0.65\,s_{\text{env}}(n))\,(0.65 + 0.35\cos\Phi(n))\,. \tag{45}$$

3. **Edge-to-Golden-Face Lerp and $90°$-per-Octant Spatial Screw:** First, the unrotated vector $\mathbf{b}_i(n)$ interpolates between the outer cube edge $\mathbf{e}_i$ and the golden Klein face center $\mathbf{g}_{f(i)}$:

$$\mathbf{b}_i(n) = r_{\text{Klein}}(n)\left[\, s_{\text{env}}(n)\,\mathbf{e}_i + (1 - s_{\text{env}}(n))\,\mathbf{g}_{f(i)}\,\right]. \tag{46}$$

   Second, the XY-plane rotates by $\theta_{\text{sp}}(n, i) = \kappa(n)\frac{\pi}{2} + 0.15\lfloor i/4 \rfloor \frac{2\pi}{3}$ (completing $720°$ of spatial rotation across the 8 octants, Proposition 2.3), while the Z-coordinate breathes with $\cos(\Phi(n)/2)$:

$$\mathbf{x}_i(n) = \begin{pmatrix} b_{i,x}(n)\cos\theta_{\text{sp}}(n,i) - b_{i,y}(n)\sin\theta_{\text{sp}}(n,i) \\ b_{i,x}(n)\sin\theta_{\text{sp}}(n,i) + b_{i,y}(n)\cos\theta_{\text{sp}}(n,i) \\ b_{i,z}(n)\cos\!\left(\frac{\Phi(n)}{2}\right) \end{pmatrix}. \tag{47}$$

5.4 Integration of Lemma 8 (Steps 3–6), Ghost Map $G(p)$, and Rational Trefoil Replacements for Section 5

Integrating Lemma 8 of *The Fifteen Lemmas* and *Golden Strassen-Clifford Toroids* into Section 5 adds four exact geometric constructions and eliminates all trigonometric calls from the trefoil and hyperbolic projection:

1. **Cube 3 Fibonacci Binomial Cube $W = [-4, 4]^3$ (Lemma 8 Step 5):** Continuing the doubling $K = [-1, 1]^3 \to V = [-2, 2]^3 \to W = [-4, 4]^3$ (edge $8C_0 = \lambda/2$, where $8 = F_6 = 5{+}3 = F_5{+}F_4$), the Golden Binomial Cube identity $n^3 = (m{+}r)^3 = m^3 {+} 3m^2r {+} 3mr^2 {+} r^3$ evaluates in exact integers to:

$$8^3 = (5 + 3)^3 = 125 + 225 + 135 + 27 = 512, \tag{48}$$

   partitioning Cube 3 into 1 major cube (5^3), 3 slabs ($3 \times 5^2 \times 3$), 3 rods ($3 \times 5 \times 3^2$), and 1 residual cube (3^3), while staying separated from T_0 by $(11 - 4\sqrt{3})C_0 \approx 4.07\,C_0 \approx 2.19$ m.

2. **The Step 6 Ghost Map $G(p) = \Phi(\mathcal{C}_{\text{rat}}(p))$ and Six Disjoint Ghost Cells (Lemma 8 Propositions 6.2 & 6.4):** For any point $\mathbf{p} \in \mathbb{Q}^3$ inside the open unit ball $|\mathbf{p}| < 1$, the Ghost Map

$G = \Phi \circ \mathcal{C}_{\text{rat}}$ maps $\mathbf{p}$ to the upper hyperboloid sheet $\mathbb{H}^3 = \{x_0^2 - |\mathbf{x}|^2 = 1, x_0 > 0\}$ via pure rational operations:

$$G(\mathbf{p}) = \left(\frac{1 + |\mathbf{p}|^2}{1 - |\mathbf{p}|^2},\ \frac{2\mathbf{p}}{1 - |\mathbf{p}|^2} \right) \in \mathbb{H}^3 \cap \mathbb{Q}^4. \tag{49}$$

Applying the six $\{4, 3, 5\}$ face-neighbour boosts $B_{\pm k}$ ($\cosh \vartheta = \varphi, \sinh \vartheta = \sqrt{\varphi}$, e.g. $B_{+1} = \begin{pmatrix} \varphi & \sqrt{\varphi} \\ \sqrt{\varphi} & \varphi \end{pmatrix} \oplus I_2$) carries the scaled trefoil $G(cT_\varphi)$ ($c = 1/12$) into the six face-neighbouring hyperbolic cubes. By Proposition 7.17 (Collapse at Step 6), at the offload window $k = 6$ (spinor $-i$), the inverse boosts B_{+k}^{-1} pull the three positive ghosts back into the central cell where $\Delta_1 \cup (-\Delta_1) = \{\pm 1\}^3$ reconstructs all 8 vertices of the core cube.

3. **Rational Weierstrass Substitution $u = \tan(t/2)$ for the φ-Trefoil (Lemma 8 Prop. 6.1 & 6.2):** In `ExactTrefoilCurve`, the curve $T_s(t) = (\sin t + 2\sin 2t,\ \cos t - 2\cos 2t,\ -s\sin 3t)$ (with $s = 1$ or $s = \varphi$) normally evaluates six transcendental sines and cosines per point. Setting $u = \tan(t/2) \in \mathbb{Q}$ gives $\cos t = \frac{1-u^2}{1+u^2}$ and $\sin t = \frac{2u}{1+u^2}$, transforming $T_s(t)$ into an **exact rational algebraic curve** over $\mathbb{Q}$ (or $\mathbb{Q}(\sqrt{5})$ for $s = \varphi$):

$$x(u) = \frac{2u}{1+u^2} + \frac{8u(1-u^2)}{(1+u^2)^2} = \frac{2u(5 - 2u^2 - 3u^4)}{(1+u^2)^3}, \tag{50}$$

$$y(u) = \frac{1-u^2}{1+u^2} - \frac{2(1 - 6u^2 + u^4)}{(1+u^2)^2} = \frac{-1 + 11u^2 - 5u^4 - u^6}{(1+u^2)^3}, \tag{51}$$

$$z(u) = -s\,\frac{2u(3-u^2)(1-3u^2)}{(1+u^2)^3}. \tag{52}$$

Furthermore, because $T_s(t)$ has D_3 dihedral symmetry of order 6, only the fundamental arc $t \in [0, \pi/3]$ ($u \in [0, 1/\sqrt{3}]$) is needed; the other 5 arcs are generated by the integer signed-permutation matrices $g_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$ and $g_2 = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \in O(3, \mathbb{Z})$.

4. **Equivalent Golden Clifford Torus Knot Inside $SU(2)$ (Lemma 8 Prop. 6.5 vs. Golden Strassen-Clifford Paper §2–3):** With the Golden Clifford Toroid radii $r_1 = \frac{\varphi}{\sqrt{\varphi+2}}$ and $r_2 = \frac{1}{\sqrt{\varphi+2}}$ ($r_1/r_2 = \varphi, r_1^2 + r_2^2 = 1$), the curve

$$K(s) = \left(r_1 e^{2is},\ r_2 e^{3is}\right) = r_1\cos 2s + r_1 \sin 2s\, i + r_2 \cos 3s\, j + r_2 \sin 3s\, k \in S^3 \tag{53}$$

is the exact $(2, 3)$ trefoil knot on the Golden Clifford Torus. As proven in Lemma 11 Part 5, $K(s)$ closes at $s = 2\pi$ because its frequency ratio $2 : 3$ is rational, whereas choosing an irrational step ratio $\Delta\theta_1/\Delta\theta_2 = \varphi$ gives the ergodic surface-scanning flow of the Golden Strassen-Clifford paper.

6 Complete Catalog of Interaction Points and Detection Functions

A primary objective of the upgraded dual-slice architecture is to trace two cubic slices—**Slice A** at voxel $\mathbf{v}_A = (v_{A,x}, v_{A,y}, v_{A,z}) \in \{-1, 0, 1\}^3$ and depth $Z_A \in [0, 1]$, and **Slice B** at adjacent voxel $\mathbf{v}_B = (v_{B,x}, v_{B,y}, v_{B,z}) \in \{-1, 0, 1\}^3$ and depth $Z_B \in [0, 1]$—and monitor every mathematical point where the two trajectories and their surrounding streams interact.

Given current frame $n \in [0, 720]$ and spinor phase offset slider $\Delta k \in \{0, \ldots, 7\}$, Slice B's effective frame index n_B is shifted by Δk macro-octants ($90\Delta k$ frames) plus Δk fine steps:

$$n_B(n, \Delta k) = (n + 90\Delta k + \Delta k) \bmod 721. \tag{54}$$

Thus Slice A is in fine spinor state $k_A = n \bmod 8$, and Slice B is in fine spinor state $k_B = (k_A + 2\Delta k) \bmod 8$ (or shifted by Δk when stepping via the interactive dial). Below are the **seven interaction mechanisms**, the exact code functions that compute them, and the Geometric Engine theorems that define them.

6.1 Interaction Point 1: Spinor Double-Cover Sheet Lock and $\pm i$ Handshake Resonance

- **Code Function:** `drawSpinorClockDial(step, stepB)` updating `#clock-dual-status`.
- **Geometric Engine Component:** Proposition 2.3(2)–(3) and Theorem 7.5 of the *Fifteen Lemmas* (Double-Cover Handshake Apices).
- **Mathematical Condition:** Let $\Delta k_{\text{fine}} = (k_B - k_A) \bmod 8$. Three distinct spinor interaction regimes occur:

$$\text{Spinor Interaction State} = \begin{cases} \textbf{APEX HANDSHAKE } (\pm i), & \text{if } \{k_A, k_B\} = \{2, 6\} \implies s_{k_A} = s_{k_B} = 1,\ q_{k_B} = \\ \textbf{SHEET LOCK } (-q), & \text{if } \Delta k_{\text{fine}} = 4 \implies \text{Rot}_x(90k_A^\circ) = \text{Rot}_x(90k_B^\circ),\ q_k \\ \textbf{PHASE SYNC } (+q), & \text{if } \Delta k_{\text{fine}} = 0 \implies q_{k_B} = +q_{k_A}. \end{cases} \tag{55}$$

  When $\{k_A, k_B\} = \{2, 6\}$, both slices simultaneously achieve maximum transverse spread $s_2 = s_6 = 1$ and identical spatial half-turn geometry $\text{diag}(1, -1, -1)$ on opposite sheets of $SU(2)$, enabling lossless onload-to-offload state transfer across adjacent cubes.

6.2 Interaction Point 2: 3D Inter-Slice Laser Bridge Resonance

- **Code Function:** `updateCompressionNodes(step)` updating `interactionBridgeLines`.
- **Geometric Engine Component:** Section 4.3 ($\{4, 3, 5\}$ Face-Neighbour Boosts) and Definition 5.3 (Lower Manifold Voxel Lattice $V = [-2C_0, 2C_0]^3$).
- **Mathematical Condition:** In 3D world space, the centers of Slice A and Slice B are positioned at:

$$\mathbf{P}_A = 25.0\ (v_{A,x},\ v_{A,y},\ v_{A,z} + Z_A - 0.5)\,, \tag{56}$$

$$\mathbf{P}_B = 25.0\ (v_{B,x},\ v_{B,y},\ v_{B,z} + Z_B - 0.5)\,. \tag{57}$$

  The 12 bridge segments connect world node $\mathbf{W}_{A,i} = \mathbf{P}_A + \mathbf{x}_i(n)$ to $\mathbf{W}_{B,i} = \mathbf{P}_B + \mathbf{x}_i(n_B)$. The 3D bridge interaction detector evaluates the local cube-frame Euclidean distance:

$$d_{\text{3D},i}(n, n_B) = \|\mathbf{x}_i(n) - \mathbf{x}_i(n_B)\|_2\,. \tag{58}$$

  Whenever $d_{\text{3D},i}(n, n_B) < 6.0$ units (i.e., within the golden Klein cell half-width $aC_0 = 0.4859 \times 12.5 = 6.073$ units!), the bridge vertex colors switch from passive cyan/magenta $(0.13, 0.83, 0.93) \to (0.91, 0.47, 0.98)$ to active golden-emerald resonance $(1.0, 0.85, 0.2) \to (0.1, 0.95, 0.6)$.

6.3 Interaction Point 3: 2D Trajectory Collision Hotspots and Resonance Rings

- **Code Function:** `drawSlicer(rotation, amp, time, step)` updating `#tele-collisions` and drawing golden collision rings.
- **Geometric Engine Component:** Proposition 3.1 (Tetrahedral Edge Squared Distance $= 8$) and Proposition 4.3.

- **Mathematical Condition:** For each of the 12 node pairs $i \in \{0, \ldots, 11\}$, `drawSlicer` computes the squared distance $d_i^2 = \|\mathbf{x}_i(n) - \mathbf{x}_i(n_B)\|_2^2$:

  1. **Hotspot Counter Threshold:** If $d_i^2 < 20.0$ units^2 ($d_i < 4.4721$ units), pair i is registered in `collisionPairs` and increments `collisionCount` (`#tele-collisions`).
  2. **2D Golden Resonance Zone:** If $d_i < 4.5$ units in the superimposed projection, a pulsing golden circle is rendered at the midpoint:

$$\mathbf{m}_i = \frac{\mathbf{x}_i(n) + \mathbf{x}_i(n_B)}{2} \cdot \text{scale}, \qquad R_{\text{pulse},i} = \max(4.0,\ 2.8\,(5.5 - d_i))\ \text{px}. \tag{59}$$

6.4 Interaction Point 4: Conformal Null-Cone Inner Product $\langle X_A, X_B \rangle$

- **Code Function:** `conformalInner` in `drawSlicer`, displayed in `#tele-nullcone`.
- **Geometric Engine Component:** Proposition 3.3 (The Conformal Null Cone in $\mathbb{R}^{4,1}$) and Axiom C2 (Linearization of Metric Constraints).
- **Mathematical Condition:** By Proposition 3.3(2), lifting a point $\mathbf{x} \in \mathbb{R}^3$ (normalized by core cube radius $C_0 = 12.5$, so $\hat{\mathbf{x}} = \mathbf{x}/C_0 \in [-1, 1]^3$) to the conformal null cone

$$X(\hat{\mathbf{x}}) = \left(\hat{\mathbf{x}},\ \frac{1}{2}\|\hat{\mathbf{x}}\|^2,\ 1\right) \in \mathbb{R}^{4,1} \tag{60}$$

with bilinear form $\langle (\mathbf{x}, \alpha, \beta), (\mathbf{y}, \alpha', \beta') \rangle = \mathbf{x} \cdot \mathbf{y} - \alpha\beta' - \beta\alpha'$ converts quadratic Euclidean distance into an exact linear inner product:

$$\langle X(\hat{\mathbf{x}}_A), X(\hat{\mathbf{x}}_B) \rangle = -\frac{1}{2}\left\|\hat{\mathbf{x}}_A - \hat{\mathbf{x}}_B\right\|_2^2. \tag{61}$$

In `drawSlicer`, the ensemble conformal null-cone coupling across all 12 node trajectories is evaluated at every frame as:

$$\langle X_A, X_B \rangle_{\text{ensemble}} = -\frac{1}{2C_0^2}\left(\frac{1}{12}\sum_{i=0}^{11}\|\mathbf{x}_i(n) - \mathbf{x}_i(n_B)\|_2^2\right). \tag{62}$$

Because $X(\hat{\mathbf{x}})$ is a null vector ($\langle X(\hat{\mathbf{x}}), X(\hat{\mathbf{x}}) \rangle = 0$), $\langle X_A, X_B \rangle_{\text{ensemble}} \leq 0$ always, reaching 0 if and only if the two slices coincide on the null cone.

6.5 Interaction Point 5: Adjacent 3×3 Cubic Lattice Energy Flux

- **Code Function:** `totalAdjFlux` and the 3×3 mini-heatmap in `drawSlicer`, displayed in `#tele-adjflux`.
- **Geometric Engine Component:** Section 6.1 & Section 6.3 (Cubic Torus Lattice $(\mathbb{Z}/m\mathbb{Z})^3$ vs. Hyperbolic Address Shadow).
- **Mathematical Condition:** For Slice A at local phase $t_A = 50Z_A - 8\tau(n) + 12v_{A,x} + 18v_{A,y} + 25v_{A,z}$, the 3×3 adjacent stream monitor samples all 9 streams $(g_x, g_y) \in \{-1, 0, +1\}^2$:

$$E_{g_x,g_y}(t_A) = |A \cdot Z(t_A + 14g_x + 21g_y)|\,, \tag{63}$$

and integrates the total adjacent boundary energy surrounding the central $[0, 0]$ core stream:

$$E_{\text{adj}}(t_A) = \sum_{\substack{g_x,g_y \in \{-1,0,1\} \\ (g_x,g_y) \neq (0,0)}} E_{g_x,g_y}(t_A). \tag{64}$$

6.6 Interaction Point 6: Tangential Bezier Vortex Coupling Between Lobes and Core

- **Code Function:** Lobe loop in `drawSlicer` and `fragmentShaderShell`.
- **Geometric Engine Component:** Proposition 3.4 (C_3 Threefold Decomposition) and Hoard (2026a) (Toroidal Orbital Stator Coupling).
- **Mathematical Condition:** In the 2D slicer, the three C_3 lobes sit at radius $R_{\text{lobe}} = 2.1 \cdot \text{scale}$ at angles $\theta_j = \frac{2\pi j}{3} + \theta_{\text{rot}}(n)$. Each lobe's instantaneous radius swells with the local Zeta energy and active soliton spikes:

$$r_j(t) = 0.6\,\text{scale}\,(1 + 0.25\,e_j(t))\,, \qquad e_j(t) = A \cdot Z(t + 20.94395j) + S(t). \tag{65}$$

  When the active CRT branch $j = n \bmod 3$ coincides with a spinor handshake apex ($s_k = 1$, i.e., $k = 2$ or $k = 6$), lobe j switches to golden apex highlight (`#f59e0b`), visually marking the exact frame where energy transfers between the rotating C_3 plane z and the central core cube.

6.7 Integration of Layer V Interaction Without Contact (Theorems 7.6–7.10) and Monoid Graph Cost into Section 6

Section 7.3 of *The Fifteen Lemmas* (*Interaction Without Contact*) and Section 4 of *Monoid Collapse and Mass-Node Traversal* add three non-contact interaction channels between engines that do not share physical points (Theorem 7.6 proves distinct engines in a stack of cell edge $> 334\,C_0$ are strictly disjoint):

1. **Interaction Point 7A — The Five-Moment Vandermonde Channel Across the 4.05 m Gap (Lemma 6 & Theorem 7.7):** Across the radial gap $r \in [2\sqrt{3}\,C_0, 11\,C_0]$ separating $\mathcal{M}_{\text{lower}}$ and $\mathcal{M}_{\text{upper}}$, let $X = r^2/(2q) \in [X_a, X_b]$ and define the five cumulative radial moments of the axial and azimuthal velocity profiles (U, E) with $H = \sqrt{2X}\,E$:

$$M(X) = \int_0^X U\,dx, \quad I(X) = \int_0^X H\,dx, \quad J(X) = \int_0^X UH\,dx, \quad S(X) = \int_0^X \left(U^2 - \frac{E^2}{2}\right) dx, \quad C_p(X) = \int \tag{66}$$

  By Lemma 6 Parts 1–4 and Theorem 7.7, the 5×5 generalized Vandermonde matrix $B_{ij} = \int_0^\infty x^{\alpha_i} \beta_j(x)\,dx$ splits into a $2 + 3$ block Jacobian that is strictly invertible whenever $\alpha \notin \{-1/2, 1/2, 3/2\}$, and Banach's contraction mapping theorem solves $Bc + Q(c, c) = d$ uniquely inside $\|c\| \le 2\|B^{-1}\|\|d\|$. Thus, five scalar numbers (M, I, J, S, C_p) at X_b completely lock the exterior stress $\mathcal{T}_0(X)$ across the gap without contact.

2. **Interaction Point 7B — The Projected Hyperbolic Quadrance Channel $Q_h(i, j)$ (Lemma 8 Prop. 6.2 & Theorem 7.8):** When Slice A has spinor state $q_A = \mathcal{C}_{\text{rat}}(\mathbf{p}_A)$ and Slice B has spinor state $q_B = \mathcal{C}_{\text{rat}}(\mathbf{p}_B)$, their non-contact interaction in $\mathbb{H}^3$ is governed by the exact rational kernel:

$$Q_h(A, B) = \langle G(\mathbf{p}_A), G(\mathbf{p}_B)\rangle_L^2 - 1 = \sinh^2 d_{\mathbb{H}}(G(\mathbf{p}_A), G(\mathbf{p}_B))\,, \tag{67}$$

  where $\langle G(\mathbf{p}), G(\mathbf{r})\rangle_L = \frac{(1+|\mathbf{p}|^2)(1+|\mathbf{r}|^2) - 4\mathbf{p}\cdot\mathbf{r}}{(1-|\mathbf{p}|^2)(1-|\mathbf{r}|^2)}$. This replaces transcendental arccosh and sinh evaluations with a single rational fraction!

3. **Interaction Point 7C — Mass-Node Relational Graph Edge Cost c_{ij} (Monoid Collapse Paper §4.2):** In *Monoid Collapse and Mass-Node Traversal*, the relational traversal cost between nodes i and j is $c_{ij} = \frac{d_{ij}^2}{M_i + M_j + \epsilon}$. Integrating Theorem 7.7 and Theorem 7.8 upgrades

d_{ij}^2 to either the conformal null-cone inner product $-2\langle X(\mathbf{x}_i), X(\mathbf{x}_j)\rangle$ (Euclidean regime) or the hyperbolic quadrance $Q_h(i,j) = \sinh^2 d_{\mathbb{H}}(G_i, G_j)$ (hyperbolic regime), while M_i, M_j are supplied by the first Vandermonde mass moment $M(X_b)$ of Lemma 6:

$$c_{ij}^{\mathsf{GET}} = \frac{Q_h(i,j)}{M_i(X_b) + M_j(X_b) + \epsilon} \geq 0. \tag{68}$$

7 64-Bit Einstein Tile Packing and the Four Parallel Tracks

In `updateQuaternionDisplay(rotation, amp, time, step)`, the live state of the center stream $[0,0,0]$ is converted into a quaternion, packed into a 64-bit Einstein Tile word (Definition 5.3), and cross-checked across the four parallel trigonometric tracks of Section 8.10.

7.1 Quaternion State Construction Along the $(1,1,1)$ Tetrahedral Axis

Given the three branch signals $a(t), b(t), c(t)$ and their C_3 apex mean $\mu_{\text{apex}} = \frac{a+b+c}{3}$, the deviation quaternion is formed and right-multiplied by the spatial rotation quaternion q_{rot} about the unit tetrahedral axis $\hat{\mathbf{u}} = \frac{1}{\sqrt{3}}(1,1,1)$:

$$q_{\text{state}} = \mu_{\text{apex}} + (a - \mu_{\text{apex}})i + (b - \mu_{\text{apex}})j + (c - \mu_{\text{apex}})k, \tag{69}$$

$$q_{\text{rot}} = \cos\left(\frac{\theta_{\text{rot}}}{2}\right) + \frac{i+j+k}{\sqrt{3}} \sin\left(\frac{\theta_{\text{rot}}}{2}\right), \qquad q_{\text{final}} = q_{\text{state}} \cdot q_{\text{rot}}. \tag{70}$$

7.2 Bit-Level Encoding into the 64-Bit Einstein Tile (Form 3 Branch Codec)

Each branch amplitude $v \in \{a, b, c\}$ is quantized into a 10-bit signed two's-complement integer in $[-511, 511]$ via `pack10Bit(v)`:

$$q_{10}(v) = \text{clamp}(\text{round}(120\, v),\, -511,\, 511) \bmod 1024 \in [0, 1023]. \tag{71}$$

Let $x_{10} = q_{10}(a)$, $y_{10} = q_{10}(b)$, and $z_{10} = q_{10}(c)$. The 64-bit Einstein Tile is assembled as two 32-bit words (`upper32` for Bits $[63:32]$ and `lower32` for Bits $[31:0]$):

$$\texttt{lower32} = ((j \;\&\; 3) \ll 30) \;\mid\; ((z_{10} \;\&\; \texttt{0x3FF}) \ll 20) \;\mid\; ((y_{10} \;\&\; \texttt{0x3FF}) \ll 10) \;\mid\; (x_{10} \;\&\; \texttt{0x3FF}), \tag{72}$$

$$\texttt{upper32} = (\texttt{0x1} \ll 30) \;\mid\; ((j \;\&\; 3) \ll 27) \;\mid\; ((k \;\&\; 7) \ll 24) \;\mid\; (\texttt{0xAE} \ll 16) \;\mid\; (\texttt{0x23} \ll 8) \;\mid\; \big(q_{10}(\mu_{\text{apex}}) \;\&\; \texttt{0xFF}\big). \tag{73}$$

Here:

- **Bits** $[29:0]$**:** Hold the three 10-bit signed branch coordinates (z_{10}, y_{10}, x_{10}).
- **Bits** $[31:30]$**:** Hold the 2-bit cyclic permutation inverse mode $j = n \bmod 3$ for $B_j^{-1} = g_3^{-j}$ (Open Problem 1).
- **Bits** $[39:32]$**:** Hold the 8-bit quantized C_3 apex invariant μ_{apex}.
- **Bits** $[47:40]$ **(`0x23`):** Nibble `0x2` selects Track B (exact primary), and nibble `0x3` specifies Form 3.
- **Bits** $[55:48]$ **(`0xAE`):** Gate byte with Holonomy Lock $= 1$ and branch validity mask $[b_2, b_1, b_0] = [1, 1, 1]$.
- **Bits** $[63:56]$**:** South Halo header encoding the 30-wheel filter class, branch $j \in \{0, 1, 2\}$, and 3-bit spinor clock step $k \in \{0..7\}$.

7.3 The Four Parallel Verification Tracks (Definition 8.10)

Finally, the Spectrum panel continuously cross-checks the four parallel trigonometric tracks of Section 8.10:

1. **Track B (Exact Primary, `#trk-b`):** Reads the exact reduced Gaussian integer state $\epsilon_k \in \{-1, 0, 1\}^2$ and rational spread $s_k \in \{0, \frac{1}{2}, 1\}$ of $(1+i)^k$, along with the exact identity $R^9 = I$ in $\mathbb{Q}(2\cos 40^\circ)$.

2. **Track A (Floating-Point Readout, `#trk-a`):** Displays the spinor angle $\alpha = 45k^\circ$, spatial angle $2\alpha = 90k^\circ$, and $\{4, 3, 5\}$ hyperbolic cell inradius $r_{\text{in}} = 0.530638$.

3. **Track C (Rational Cayley Approximant, `#trk-c`):** Applies the rational quaternion $q_C = 138 + 29(i + j + k)$ of norm $N = 138^2 + 3(29^2) = 21,567$ (Proposition 8.7 and Table 13), which approximates the 40° wheel rotation to within 0.00116° while preserving the circumsphere quadrance $\|\mathbf{x}\|^2 = 3.000000$ exactly over $\mathbb{Q}$.

4. **Track R (Triplet Prime Residue Fingerprint, `#trk-r`):** Because $p = 881$ satisfies $881 \equiv 1 \pmod 4$, $881 \equiv 1 \pmod 5$, and $881 \equiv -1 \pmod 9$, it splits $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{5})$, and $\mathbb{Q}(2\cos 40^\circ)$ simultaneously into 10-bit integer arithmetic ($i \mapsto 387$, $\sqrt{5} \mapsto 228$, $2\cos 40^\circ \mapsto 120, 302, 459 \pmod{881}$, Proposition 8.8). The code computes the modular quadrance fingerprint:

$$Q_{881} \equiv \left(x_{10}^2 + y_{10}^2 + z_{10}^2\right) \pmod{881}, \tag{74}$$

   certifying the integer payload in $O(1)$ word operations within the Form 2 Garner CRT range $M = 881 \times 883 \times 887 = 690,017,701$.

7.4 Integration of Section 6.1 (Forms 0–15 Codex Shapes), Lemma 9 Closed-Form Garner Inverses, and Lemma 15 Merkle Security into Section 6

Integrating Section 6.1, Lemma 9, and Lemma 15 of *The Fifteen Lemmas* completes the bit-level protocol specification:

1. **Exact Closed-Form Polynomial Garner Inverses c_{12}, c_{23}, c_{13} (Lemma 9 Part 2):** For any prime triplet $(p, p+2, p+6)$ with $p > 5$ (which must lie in $(11, 13, 17) \pmod{30}$ or $(17, 19, 23) \pmod{30}$ by Lemma 9 Part 1), reconstructing $X \in [0, M)$ from its residues (x_1, x_2, x_3) via

$$u_1 = x_1, \quad u_2 = (x_2 - u_1)c_{12} \bmod (p+2), \quad u_3 = ((x_3 - u_1)c_{13} - u_2)c_{23} \bmod (p+6), \quad X = u_1 + u_2 p + u_3 p(p+2) \tag{75}$$

   requires zero extended-GCD calls or lookup tables because all three modular inverses are exact closed-form polynomials in p:

$$c_{12} = \frac{p+1}{2}, \qquad c_{23} = \begin{cases} \frac{3p+17}{4} & p \equiv 1 \pmod 4, \\ \frac{p+5}{4} & p \equiv 3 \pmod 4, \end{cases} \qquad c_{13} = \begin{cases} \frac{p+5}{6} & p \equiv 1 \pmod 6, \\ \frac{5p+29}{6} & p \equiv 5 \pmod 6. \end{cases} \tag{76}$$

   Evaluating at $p = 881$ ($881 \equiv 1 \pmod 4$ and $881 \equiv 5 \pmod 6$) yields the exact constants $c_{12} = 441$, $c_{23} = 665$, and $c_{13} = 739$.

2. **Complete Form ID Table (Forms 0–15) and Codex Shapes (Section 6.1, Tables 4–6):** In the canonical Fifteen Lemmas bit layout (Table 4 of *Fifteen Lemmas*), Bits $[63 : 61]$ hold the 3-bit cycle step k, Bits $[60 : 58]$ hold the 3-bit wheel unit $u \in \{0..7\}$, Bits $[57 : 56]$ hold the 2-bit structure selector ($I, A, \bar{A}, E$ or $\mathcal{M}^{(1)}..\mathcal{M}^{(4)}$), Bits $[55 : 48]$ hold the Gate Byte, Bits $[47 : 44]$ hold the step-multiplexed index ($s \in \mathbb{Z}/9$, half-cube $l \in \{0..15\}$, ghost index $0..6$, or $\mathfrak{aff}(3)$ coordinate), Bits $[43 : 40]$ hold the 4-bit Form ID ($0..15$), Bits $[39 : 36]$ hold the stream slot ($0..15$, forming the 6-bit Stream ID with Bits $[57 : 56]$), and Bits $[35 : 32]$ hold the Scale

Register ($2^l, F_m$, or geodesic frequency ν). Forms 4–15 name deterministic geometric codex shapes: Form 4 (64-cell Voxel Cube V), Form 5 (8-cell Core K), Form 6 (Stella Octangula $\Delta_1 \cup -\Delta_1$), Form 7 (12-point Cuboctahedron), Form 8 (Geodesic Icosahedron of $10\nu^2+2$ points), Form 9 (24 Hurwitz Units), Form 10 (12-element Binary Dihedral Group $2D_3$), Form 11 (9-Ray Circle $x(s)$), Form 12 (7 Hyperbolic $\{4,3,5\}$ Cells), Form 13 (F_m Golden Circle Points), Form 14 (8 Wheel Units mod 30), and Form 15 (Merkle-attested user codex).

3. **Recursive Bottom-Up Merkle Attestation vs. South Halo Cache Pre-Filter (Lemma 15):** Because a 12-layer 64-voxel state has $64^{12} = 2^{72}$ states (≥ 256 preimages per 64-bit word) and the 8-bit South Halo has $2^{56}/2^8 = 2^{48}$ payloads per header ID, Lemma 15 proves that the South Halo is strictly an $O(1)$ L1/L2 cache pre-filter, whereas full structural integrity is guaranteed by the bottom-up recursive Merkle chain $\tau_{12}(G) = H(G_{12})$, $\tau_d(G) = H(G_d \parallel \tau_{d+1}(G))$ ($d = 11, \ldots, 1$) using a 256-bit collision-resistant hash (SHA-256/BLAKE3, 2^{128} birthday security).

8 Summary of Interaction Thresholds and Reference Parameters

Table 3 consolidates all numerical constants, geometric radii, and interaction thresholds active in `zeta_energy_system.html`.

Table 3: Summary of exact geometric constants and interaction thresholds in `zeta_energy_system.html`.

Parameter / Threshold	Symbol	Exact / Numerical Value	Governing Proposition
Voxel Lattice Spacing	C_{space}	25.0 units	Def. 5.3 ($V = [-2C_0, 2C_0]^3$)
Core Cube Half-Width	C_0	12.5 units	Def. 5.3 ($K = [-C_0, C_0]^3$)
$\{4,3,5\}$ Klein Cell Half-Width	a	$\varphi^{-3/2} \approx 0.48586827$	Prop. 4.3 ($a^2 = \sqrt{5}-2$)
$\{4,3,5\}$ Cell Physical Half-Width	aC_0	6.07335 units	Prop. 4.3
$\{4,3,5\}$ Cell Inradius	r_{in}	$\operatorname{artanh}(a) \approx 0.53063753$	Prop. 4.3(2), Prop. 8.4(1)
Face-Neighbour Boost Rapidity	ϑ	$2r_{\text{in}} \approx 1.06127506$	Prop. 4.3(2) ($\cosh\vartheta = \varphi$)
Scaled Trefoil Max Klein Radius	r_{trefoil}	$0.2520 < a$	Section 4.3, Table 18 #7
3D Inter-Slice Bridge Threshold	$d_{\text{3D,max}}$	6.0 units $\approx aC_0$	Section 5.2 (Bridge Resonan
3D Collision Hotspot Threshold	d^2_{hotspot}	20.0 units^2 ($d < 4.472$ units)	Section 5.3 (`#tele-collisio`
2D Golden Resonance Ring Cutoff	d_{ring}	4.5 units	Section 5.3 (`drawSlicer`)
Same-Branch Handshake Delay	Δ_{CRT}	12 frames ($\Delta \equiv 4 \bmod 8$)	Prop. 3.7(3)
Master Multi-Clock Closure	N_{sync}	$\operatorname{lcm}(24,2,3,5) = 120$ frames	Prop. 3.8, Table 11
Triplet Prime CRT Modulus	M	$881 \times 883 \times 887 = 690,017,701$	Axiom C5, Prop. 8.8

8.1 Integration of the Unified Metrological Standardization Table (Section 6.2 of the Fifteen Lemmas) into Section 7

At the standard calibration frequency $f_{\text{cal}} = 40$ Hz in air at $20°$C, 50% relative humidity, and 101.325 kPa ($v_{\text{local}} = 343.98$ m/s, wavelength $\lambda = 8.5995$ m), every simulation threshold in Table 3 translates directly into physical SI dimensions via $1\,C_0 = \lambda/16 = 53.748$ cm $\pm\, 0.022$ cm ($dC_0/dT \approx 0.98$ mm/K):

- **Four-Fold $1/16$ Half-Cube Footprint (Lemma 8 Step 3):** $\frac{1}{16}C_0 = \frac{\lambda}{256} = 3.359$ cm (acoustic phase $\pi/128$).
- **Harmonic Offset $MF = x_F - x_M$ (Lemma 5 Part 3):** $\frac{1}{3}C_0 = \frac{\lambda}{48} = 17.916$ cm (phase $\pi/24$).

- **Centered Radial Harmonic Conjugate F_{rad} (Lemma 5 Part 3):** $\frac{2}{3}C_0 = \frac{\lambda}{24} = 35.832$ cm (phase $\pi/12$).
- **Klein-Wave Merge Gate x_F (Lemma 5 Part 3):** $\frac{4}{3}C_0 = \frac{\lambda}{12} = 71.664$ cm (phase $\pi/6$).
- **Apollonius Circle Radius $R_A = 2\sqrt{2}\,C_0$ (Lemma 5 Part 3b):** $\frac{\sqrt{2}}{8}\lambda = 1.5202$ m.
- **Cross-Manifold Separation Gap** $\text{dist}(V, T_0) = (11-2\sqrt{3})C_0$ **(Lemma 10 Part 3):** $7.5359\,C_0 = 0.471\lambda = 4.0504$ m.
- **Outer Engine Enclosure Radius** $(R_4 + r_4)C_0 = 167\,C_0$ **(Lemma 10 Part 3):** $\frac{167}{16}\lambda = 89.76$ m.

9 Layer V System-Level Integration: The 3:1 FCC Interleave, 12-Engine Memory, Four-Structure Sandwich, and 64 Streams

Section 7 of *The Fifteen Lemmas* derives, without introducing any new lemma, how multiple Geometric Engines stack and operate together as a fault-tolerant, angle-conserving computational fabric. Integrating Layer V into the Zeta Energy System connects the $3 \times 3 \times 3$ cubic grid to four system-level theorems:

9.1 The 3:1 Face-Centered Cubic (FCC) Interleave and the Implied Fourth (Theorems 7.1–7.3)

Let $L = 2\mathbb{Z}^3$ and define the four Klein four-group offsets $o_0 = (0,0,0)$, $o_1 = (1,1,0)$, $o_2 = (1,0,1)$, and $o_3 = (0,1,1)$. By Theorem 7.2:

1. The four disjoint translates $L + o_i$ partition the face-centered cubic (FCC) lattice $D_3 = \{x \in \mathbb{Z}^3 : x_1 + x_2 + x_3 \text{ even}\}$.
2. The 120° rotation g_3 about $(1,1,1)$ fixes $L + o_0$ and cycles the three face translates $L + o_1 \to L + o_3 \to L + o_2$.
3. Since $o_1 + o_2 + o_3 \equiv (0,0,0) \pmod 2$, any three translates uniquely determine the **Implied Fourth**—the exact lattice twin of the quaternion group inverse $H_4 = (H_1H_2H_3)^{-1}$ (Lemma 7 Part 3b) and the quaternion basis identity $ijk = -1$.
4. Each point of D_3 has 12 nearest neighbours—the cuboctahedron vectors $(\pm1, \pm1, 0)$ and their permutations (Codex Form 7), which split $3+6+3$ along $(1,1,1)$ and partition by their zero coordinate into three planar groups of four summing to zero (Proposition 7.13).

9.2 The Twelve-Engine Parity Hierarchy and the 13th Meta-Parity Engine (Propositions 7.11–7.14)

In the $3 \times 3 \times 3$ cubic array surrounding the center stream $[0,0,0]$, the 12 edge-midpoint cuboctahedral neighbours form the **Twelve-Engine Hierarchy**:

$$\text{Group } A: \; F_A = (A_1A_2A_3)^{-1}, \qquad \text{Group } B: \; F_B = (B_1B_2B_3)^{-1}, \qquad \text{Group } C: \; F_C = (C_1C_2C_3)^{-1}, \tag{77}$$

comprising 9 data engines (75%) and 3 implied fourth parity engines (F_A, F_B, F_C). Adding the central $[0,0,0]$ engine as the **13th Engine** $F_* = (F_AF_BF_C)^{-1}$ (the fourth of the fourths, Proposition 7.11) allows the array to survive the simultaneous loss of any data engine *together with its own group's parity fourth* (e.g., A_1 and F_A), because F_*, F_B, F_C first reconstruct $F_A = (F_BF_CF_*)^{-1}$, which then reconstructs $A_1 = (A_2A_3F_A)^{-1}$! By Proposition 7.14, routing the implied fourths' traffic in the third 120° time slot leaves the first two phases purely for data, achieving a $4/3\times$ data-rate gain (576 data bits + 192 parity bits).

9.3 The Four-Structure Sandwich System $(I, A, \bar{A}, E)$ and 64 Logged Streams (Theorems 7.18–7.20)

Section 7.6 and 7.7 of *The Fifteen Lemmas* organize four 12-engine rings into the **Four-Structure Sandwich** encoded by Bits $[57 : 56]$ of the Einstein Tile:

- $\texttt{00} = I$: Input 12-engine structure ($I_1, \ldots, I_{12}$),
- $\texttt{01} = A$: Forward Balancer structure ($A_1, \ldots, A_{12}$),
- $\texttt{10} = \bar{A}$: Inverse Balancer structure ($\bar{A}_j = A_j^{-1}$),
- $\texttt{11} = E$: Emergent structure fixed by the slot-wise sandwich closure:

$$A_j I_j A_j^{-1} E_j = 1 \iff E_j = A_j I_j^{-1} A_j^{-1}, \qquad j = 1, \ldots, 12. \tag{78}$$

This sandwich geometry proves three fundamental theorems:

1. **Theorem 7.18 (Balancers Conserve Rotation Angles):** Because conjugation preserves the scalar real part of any quaternion, $\mathrm{Re}(E_j) = \mathrm{Re}(I_j^{-1}) = \mathrm{Re}(I_j)$, so the emergent structure E has the *exact same rotation angles* as the input I, about axes rotated by A_j.
2. **Theorem 7.19 (Nested Cancellation Law):** Forward and inverse quaternion streams cancel identically for all factor values if and only if they are nested palindromically: $q_1 \cdots q_g q_g^{-1} \cdots q_1^{-1} = 1$, whereas non-nested $q_1 \cdots q_g q_1^{-1} \cdots q_g^{-1} \neq 1$.
3. **Proposition 7.20 (Automatic Group Closure & 3-of-8 Erasure Recovery):** Whenever $I_1 I_2 I_3 I_4 = 1$, the emergent group closes automatically in reverse order: $E_4 E_3 E_2 E_1 = A(I_1 I_2 I_3 I_4)^{-1} A^{-1} = 1$. Every loss of up to 3 of the 8 engines in a paired group (and 64 of the 70 possible 4-engine losses) is recovered bit-exactly!

Together, the 4 structures $\times$ (12 boundary engine streams + 4 interior projection streams w, x, y, z) = **64 Streams** logging $64 \times 64 = 4,096$ bits per cycle step across the $4 \times 4 \times 4$ voxel cube V.

10 Integration of Monoid Collapse and Mass-Node Traversal in Relational Space

The companion report *Monoid Collapse and Mass-Node Traversal in Relational Space: A Hybrid Coarse-Grained Search Framework* formalizes the transition from continuous 3D volumetric diffusion $\partial\rho/\partial t = D\nabla^2\rho$—where the Gaussian point-source kernel $\rho(r,t) = (4\pi Dt)^{-3/2} \exp(-r^2/4Dt)$ spreads as $r(t) \sim \sqrt{Dt}$ and requires evaluating $N_{\text{grid}} \sim (R/h)^3$ grid cells—to a two-stage hybrid search architecture:

1. **Stage 1 (Algebraic Monoid Homomorphism):** A map $\phi : X \to (M, \cdot, e)$ satisfying $\phi(T_1 \circ T_2) = \phi(T_1) \cdot \phi(T_2)$, compressing a k-step transformation chain $T = T_1 \circ \cdots \circ T_k$ into an associative product $\phi(T) = \phi(T_1) \cdots \phi(T_k)$.
2. **Stage 2 (Sparse Mass-Node Graph Traversal):** A nonnegative weighted graph $G = (V, E, c)$ with gravitational-style edge cost

$$c_{ij} = \frac{d_{ij}^2}{M_i + M_j + \epsilon} \geq 0 \qquad (\epsilon > 0), \tag{79}$$

solved for the optimal path $P^* = \arg\min_{P: s \to t} \sum_{(i,j) \in P} c_{ij}$ via Dijkstra's algorithm in $O((|V| + |E|) \log |V|)$ time whenever $|V| \ll (R/h)^3$ (Theorem 5.1).

10.1 Resolving the Open Conjectures of the Monoid Collapse Paper Using the Geometric Engine

In Section 6 (Table 1) of the Monoid Collapse report, two premises were left as `[HEURISTIC]` or `[CONJECTURAL]`: (a) constructing an explicit, bounded-cost monoid homomorphism $\phi : X \to M$ that enables fast jump-ahead without bit-width explosion, and (b) grounding the heuristic cost function c_{ij} in exact Engine invariants. *Hoard (2026i)* and *The Fifteen Lemmas* resolve both:

Table 4: Explicit realizations of the Monoid Homomorphism $\phi : X \to (M, \cdot, e)$ supplied by the Geometric Engine.

Monoid $(M, \cdot, e)$	**Homomorphism** $\phi(T)$	**Jump-Ahead Complexity**	**Governing Lemma / Theorem**
1. Projective Binary Octahedral Group $M = 2O/\mathbb{R}^\times$ ($\lvert M\rvert = 48$)	Maps cube rotations and $(1+i)^k$ spinor steps to $\epsilon_k \in \{-1, 0, 1\}^2$ and $2T$.	**$O(1)$ word ops** ($g^N = g^{N \bmod n}$, zero bit growth).	Lemma 5 Part 2; Lemma 13 Part 2 & 4; Thm. 5.1(1)
2. Joint CRT Schedule Ring $M = \mathbb{Z}/8\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$	$\phi(n) = (n \bmod 8,\ n \bmod 3) \in \mathbb{Z}/24\mathbb{Z}$.	**$O(1)$ modular addition**; locks onload/offload at $\Delta = 12$.	Hoard (2026i) Def. 3.6 & Prop. 3.7
3. Right-Acting Pump Holonomy Monoid $M = S^3 \cap \mathbb{Q}^4$	$\phi(T_1 \circ \cdots \circ T_{720}) = Q^{(8)}Q^{(10)}Q^{(12)} = q_0 q_{719}^{-1}$.	**$O(1)$ endpoints** for coboundaries; **$O(\log k)$ depth** via prefix scan.	Lemma 12 Parts 1–3; Hoard (2026b) Prop. 2
4. Triplet Prime CRT Ring $M = \mathbb{Z}/881 \times \mathbb{Z}/883 \times \mathbb{Z}/887$	$\phi(Q) = (Q \bmod 881, Q \bmod 883, Q \bmod 887)$.	**$O(\log N)$ square-and-multiply** in 30-bit words; $O(1)$ Garner lift.	Lemma 9 Part 2; Hoard (2026i) Prop. 8.8

Furthermore, in the Zeta Energy System's $3 \times 3 \times 3$ lattice (and the 12-engine hierarchy of Layer V), the mass-node graph $G = (V, E, c)$ is instantiated directly on the 27 voxel streams (or 12 cuboctahedral engines), where:

$$d_{ij}^2 = \begin{cases} -2C_0^2 \langle X(\hat{\mathbf{x}}_i), X(\hat{\mathbf{x}}_j) \rangle & \text{(Conformal } \mathbb{R}^{4,1} \text{ Euclidean quadrance, Lemma 8 Step 2)}, \\ Q_h(i,j) = \langle G(\mathbf{p}_i), G(\mathbf{p}_j) \rangle_L^2 - 1 & \text{(Hyperbolic } \mathbb{H}^3 \text{ Ghost quadrance, Theorem 7.8)}, \end{cases} \tag{80}$$

and the node mass $M_i = M_i(X_b) = \int_0^{X_b} U_i(x)\, dx$ is the conserved first Vandermonde moment of Lemma 6 (or the local C_3 apex energy $s_i^2 = (a_i + b_i + c_i)^2$).

11 Integration of Golden Strassen-Clifford Toroids and Sub-Cubic 16-MAC Tensor Execution

The fourth paper, *Quaternionic Delta Encoding on Golden Strassen-Clifford Toroids: Wave Collision and Sub-Cubic Matrix Optimization in the SHD-CCP 64-Bit Protocol*, introduces two major components: (1) mapping differential quaternion updates $q_{n+1} = \Delta q_n \cdot q_n$ ($\Delta q = e^{i\Delta\theta_1} e^{j\Delta\theta_2}$) onto the

Golden Clifford Torus in $S^3 \subset \mathbb{R}^4$:

$$x^2 + y^2 = r_1^2 = \frac{\varphi^2}{1+\varphi^2} = \frac{\varphi^2}{\varphi+2}, \qquad z^2 + w^2 = r_2^2 = \frac{1}{1+\varphi^2} = \frac{1}{\varphi+2}, \qquad \frac{r_1}{r_2} = \varphi, \tag{81}$$

and (2) exploiting the structured 4×4 matrix representation of quaternion multiplication to reduce naive 4×4 matrix-matrix multiplication from 64 scalar multiplications ($O(N^3)$) down to **16 core multiply-accumulate (MAC) operations** per state step.

11.1 How the Fifteen Lemmas Rigorize and Optimize the Golden Strassen-Clifford Toroid

1. **Scalar-Plus-Skew-Symmetric 4×4 Matrix Decomposition (Lemma 8 Step 2.1):** Lemma 8 Step 2.1 proves why the 4×4 left-multiplication matrix $Q_a \in M_4(\mathbb{R})$ of a quaternion $q_a = w_a + x_a i + y_a j + z_a k$ requires only 16 scalar multiplications when multiplying another quaternion state (or 4×4 block):

$$Q_a = w_a I_4 + S(x_a, y_a, z_a) = \begin{pmatrix} w_a & -x_a & -y_a & -z_a \\ x_a & w_a & -z_a & y_a \\ y_a & z_a & w_a & -x_a \\ z_a & -y_a & x_a & w_a \end{pmatrix}, \qquad Q_a^T Q_a = N(q_a)\, I_4, \qquad \det Q_a = N(q_a)^2. \tag{82}$$

   Because Q_a is completely determined by the 4 parameters (w_a, x_a, y_a, z_a) rather than 16 independent matrix entries, multiplying $Q_a \cdot q_b$ executes as a 4-input, 16-MAC directed acyclic graph (DAG). Even better, when Δq_a is one of the 8 reduced Gaussian clock states $\epsilon_k \in \{-1, 0, 1\}^2$ (Lemma 13 Part 4) or one of the signed-permutation matrices $U_{\text{even}}, U_{\text{odd}}, g_3, g_2 \in O(3, \mathbb{Z})$ (Lemma 4 Part 2 & Lemma 8 Step 6), all entries lie in $\{-1, 0, +1\}$, reducing the 16 multiplications to **zero multiplications** (pure bit-sign flips and register swaps)!

2. **Replacing Trigonometric Toroidal Coordinates $q(\theta_1, \theta_2)$ with Rational Cayley & Lucas/Fibonacci Tracks:** In §3.1 of the Golden Strassen-Clifford paper, a point on the toroid is parameterized trigonometrically as:

$$q(\theta_1, \theta_2) = (r_1 \cos\theta_1) + (r_1 \sin\theta_1) i + (r_2 \cos\theta_2) j + (r_2 \sin\theta_2) k. \tag{83}$$

   Using the Exact-First Convention (Hoard 2026i Section 8) and Lemma 8 Proposition 6.5, we replace $\cos\theta_1, \sin\theta_1, \cos\theta_2, \sin\theta_2$ with two rational half-angle parameters $u_1 = \tan(\theta_1/2) \in \mathbb{Q}$ and $u_2 = \tan(\theta_2/2) \in \mathbb{Q}$:

$$q(u_1, u_2) = r_1 \left(\frac{1 - u_1^2}{1 + u_1^2} + \frac{2u_1}{1 + u_1^2} i \right) + r_2 \left(\frac{1 - u_2^2}{1 + u_2^2} j + \frac{2u_2}{1 + u_2^2} k \right), \tag{84}$$

   which satisfies $N(q(u_1, u_2)) = r_1^2 + r_2^2 = \frac{\varphi^2+1}{\varphi+2} = 1$ identically for all $u_1, u_2 \in \mathbb{Q}$ without a single sine, cosine, or floating-point drift!

12 Master Compendium of $\sin, \cos, \tan, \sinh, \cosh, \tanh, \exp, \log$ Replacement Formulas and Optimizations

To make the entire mathematical toolkit directly usable across shaders, hardware pipelines, and analytical proofs, Table 5 and Equations (85)–(88) catalog every transcendental replacement formula and algorithmic optimization established across the four monographs.

Table 5: Master lookup table of transcendental functions replaced by exact algebraic, rational, or integer operations in the Geometric Engine.

Legacy Function	**Where It Appeared**	**Exact Geometric Engine Replacement**	**Source Lemma / Prop.**
$\cos(k\pi/4)$, $\sin(k\pi/4)$, $\tan(k\pi/4)$	8-step spinor clock & transverse envelope $A_k = \lvert\sin(k\pi/4)\rvert$	Projective Gaussian integer $(1+i)^k/2^{\lfloor k/2\rfloor} = a_k + ib_k \in \{-1,0,1\}^2$; rational spread $s_k = \frac{b_k^2}{a_k^2+b_k^2} \in \{0, \frac{1}{2}, 1\}$; $\tan^2 = \frac{b_k^2}{a_k^2}$.	Lemma 13 Part 4; Hoard (2026i) Prop. 2.2, 8.2
$\cos\theta, \sin\theta, \tan\theta$ (general angle)	Unit circle, S^3 rotations, Golden Clifford Toroid $q(\theta_1, \theta_2)$	Rational Cayley / Weierstrass parameter $u = \tan(\theta/2) \in \mathbb{Q}$: $\cos\theta = \frac{1-u^2}{1+u^2}$, $\sin\theta = \frac{2u}{1+u^2}$, $\tan\theta = \frac{2u}{1-u^2}$.	Lemma 9 Part 3; Lemma 8 Prop. 6.2
$\sin t + 2\sin 2t$, $\cos t - 2\cos 2t$, $-s\sin 3t$	Parametric $(2,3)$-trefoil knot $T_s(t)$ in 3D and $\mathbb{H}^3$	Rational degree-6 polynomials in $u = \tan(t/2)$ over $[0, \pi/3]$ + 6 integer D_3 matrices $g_3, g_2 \in O(3,\mathbb{Z})$.	Lemma 8 Step 6 & Prop. 6.1–6.2
$\cos 40^\circ, \sin 40^\circ$	9-phase wheel μ_9 & tetrahedra Δ_1, Δ_2	Cubic field $\mathbb{Q}(c)$ ($c^3 - 3c + 1 = 0$) circulant $R(\rho_0, \rho_1, \rho_2)$ ($9y^3 - 9y^2 + 1 = 0$); mod 881: $c \equiv 120, 302, 459$.	Lemma 8.2; Hoard (2026i) Prop. 8.3, 8.8
$\frac{\sin\angle BAD}{\sin\angle CAD}$, $\frac{\sin C}{\sin B}$	Triangle bisectors, symmedians, & Klein-wave gate x_F	Symmedian & Apollonius quadrance ratio $\frac{BF}{FC} = \frac{EB}{EC} = \frac{Q_3}{Q_2}$; harmonic bundle $(0, 2C_0; 4C_0, \frac{4}{3}C_0) = -1$.	Lemma 3 Parts 1–3; Lemma 5 Parts 1, 3
$\lVert u\rVert\lVert v\rVert\sin\theta$, $\sqrt{s(s-a)(s-b)(s-c)}$	Triangle area, cross products, & Weitzenböck χ	Two-product 128-bit integer Gram quadrea $A_R = 4(Q_2Q_3 - \langle u,v\rangle^2) < 2^{86}$; $\sin^2\theta = \frac{A_R}{4Q_2Q_3}$.	Lemma 4 Part 2; Lemma 5 Part 6(a)
$\cos(N\theta), \sin^2(N\theta)$	Multi-step infinite-order rotations $(3+4i)^N/5^N$	Chebyshev $T_N(3/5) = a_N/5^N$ and Spread Polynomial recursion $S_N(s) = 2(1-2s)S_{N-1} - S_{N-2} + 2s$.	Lemma 13 Part 1; Hoard (2026i) Prop. 8.6
$\theta_1 + \theta_2 + \theta_3 = \pi$	Euclidean flatness vs. hyperbolic area defect δ	Triple Spread Formula $(s_1+s_2+s_3)^2 - 2(s_1^2+s_2^2+s_3^2) - 4s_1s_2s_3 = 0$ (Euclidean) or $\sin\delta\prod\sin(\cdots)$.	Hoard (2026i) Prop. 8.9
$\cosh d_{\mathbb{H}}, \sinh d_{\mathbb{H}}$, $\operatorname{artanh}\lvert\mathbf{p}\rvert$	Hyperbolic distance & non-contact kernel in $\mathbb{H}^3$	Ghost map $G(\mathbf{p}) = \left(\frac{1+\lvert\mathbf{p}\rvert^2}{1-\lvert\mathbf{p}\rvert^2}, \frac{2\mathbf{p}}{1-\lvert\mathbf{p}\rvert^2}\right)$ and rational kernel $Q_h(i,j) = \langle G_i, G_j\rangle_L^2 - 1 = \sinh^2 d_{\mathbb{H}}$.	Lemma 8 Prop. 6.2; Theorem 7.8

Legacy Function	Where It Appeared	Exact Geometric Engine Replacement	Source Lemma / Prop.
$\cosh\vartheta, \sinh\vartheta, \tanh(\vartheta/2)$	$\{4,3,5\}$ honeycomb boosts & Fibonacci torus boosts	Exact golden algebraic ring $\mathbb{Z}[\sqrt{\varphi}]$ ($\cosh\vartheta = \varphi, \sinh\vartheta = \sqrt{\varphi}, \tanh\frac{\vartheta}{2} = \varphi^{-3/2}$) & Lucas integers $L_n^2 - 5F_n^2 = 4(-1)^n$.	Lemma 8 Prop. 6.4; Hoard (2026i) Prop. 8.4, 8.5
$q^w = \exp(w\log q)$, $\epsilon = \log H$	Torsional Pump upward SLERP & holonomy correction	Rational Cayley tangent scaling $\mathcal{C}_{\text{rat}}(\mathbf{u}_w)$ and BCH cross-product series $-\log H_4 = \sum\epsilon_a + \sum_{a<b}\epsilon_a \times \epsilon_b$.	Lemma 7 Part 3(b); Lemma 9 Part 3
$e^{-\beta(V(b)-V(a))}$	Markov pump & lattice transition probabilities	Rational valve kernel $\rho_e = \frac{1}{1+\lambda A_e}$ and Sparsemax projection on Q_h logits.	Lemma 7 Part 3(c), 3(e); Theorem 7.9

12.1 Explicit Algebraic Replacement Equations for Direct Implementation

1. Complete Rational Substitution for $\sin, \cos, \tan$ **and 3D Rotations (Lemma 9 Part 3):** For any rational tangent half-angle vector $\mathbf{u} = (u_x, u_y, u_z) = \tan(\theta/2)\hat{\mathbf{u}} \in \mathbb{Q}^3$ with quadrance $Q_u = \|\mathbf{u}\|^2 = u_x^2 + u_y^2 + u_z^2$:

$$\cos\theta = \frac{1-Q_u}{1+Q_u}, \qquad \sin^2\theta = \frac{4Q_u}{(1+Q_u)^2}, \qquad \tan^2\theta = \frac{4Q_u}{(1-Q_u)^2}, \tag{85}$$

and the exact rational $SO(3,\mathbb{Q})$ rotation matrix acting on $\mathbf{x} \in \mathbb{Q}^3$ without square roots is:

$$\text{Rot}(\mathbf{u})\mathbf{x} = \mathbf{x} + \frac{2}{1+Q_u}\left(\mathbf{u}\times\mathbf{x} + \mathbf{u}\times(\mathbf{u}\times\mathbf{x})\right). \tag{86}$$

2. Exact Symmedian and Harmonic Bundle Replacement for Sine Ratios (Lemma 3 & Lemma 5): Given triangle vertex A and baseline $BC = [0, 2C_0]$ with side quadrances $Q_3 = \|A-B\|^2$ and $Q_2 = \|A-C\|^2$, the internal symmedian foot x_F and external Apollonius tangent x_E are computed with zero trigonometry:

$$x_F = \left(\frac{Q_3}{Q_2+Q_3}\right)2C_0, \qquad x_E = \left(\frac{Q_3}{Q_3-Q_2}\right)2C_0, \qquad (0, 2C_0; x_E, x_F) = -1. \tag{87}$$

3. Exact Hyperbolic Ghost Kernel and BCH Logarithm Replacement (Lemma 7 & Lemma 8): For any two Poincaré ball states $\mathbf{p}, \mathbf{r} \in \mathbb{Q}^3$ ($|\mathbf{p}|, |\mathbf{r}| < 1$) and three small drift vectors $\epsilon_1, \epsilon_2, \epsilon_3 \in \mathbb{Q}^3$:

$$Q_h(\mathbf{p},\mathbf{r}) = \left(\frac{(1+|\mathbf{p}|^2)(1+|\mathbf{r}|^2) - 4\mathbf{p}\cdot\mathbf{r}}{(1-|\mathbf{p}|^2)(1-|\mathbf{r}|^2)}\right)^2 - 1, \qquad \epsilon_{\text{net}} = \epsilon_1+\epsilon_2+\epsilon_3+\epsilon_1\times\epsilon_2+\epsilon_1\times\epsilon_3+\epsilon_2\times\epsilon_3. \tag{88}$$